

Dendritic morphology as a dictionary for local credit assignment

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Abstract

Dendritic trees compartmentalize synaptic inputs, but which spatial distinctions a learning signal must preserve is unclear. We describe delivered credit as combinations of spatial profiles constrained by morphology. In image classifiers with one dendritic neuronal layer, preserving neuronal identity accounts for most of the feedback benefit. Contextual conflict favors branch selection, while hierarchical distractors favor ancestry grouping at a bandwidth predicted by the task. Targets with matched input-sensitivity spectra develop different learned credit spectra and tolerance of fixed profiles; a local inhibitory gate also supports learning of opposing branch tuning in a conductance tree. Reconstructed arbors compress modeled passive perturbation fields through ancestry profiles, and focal shunts regulate their gains with selectivity set by electrotonic state. Measured visual responses provide no evidence of preferential ancestry alignment in the sampled cells. The framework links task structure to useful credit distinctions while separating available delivery patterns from the mechanisms that generate their coefficients.

Introduction

Learning must link a synapse’s local activity to its contribution to a circuit outcome. Cable filtering, nonlinear integration and spatially organized inhibition make locations within a dendritic tree functionally distinct [1–7]. These differences expand computation, but when does branching also help distribute useful learning signals?

Credit assignment asks how changing a synapse would affect task error. Backpropagation (BP) answers this by calculating the loss gradient—the sensitivity of task error to each parameter [8]; related sensitivity calculations apply to recurrent and steady-state networks [9, 10]. For a point neuron, each incoming-weight gradient is the product of a locally available sensitivity, called eligibility, and a neuron-specific learning signal. Hebbian plasticity captures the association between local pre- and postsynaptic variables [11], while three-factor rules combine that eligibility with an outcome-dependent signal [12]. Here, a rule is local when a synapse uses its own eligibility and an available learning signal. This definition does not require that the signal itself be computed locally.

Biologically motivated learning theories differ in how they obtain such signals. Feedback alignment communicates errors through separate pathways [13, 14]; equilibrium, predictive and prospective methods infer target states or errors from dynamics [15–17]; eligibility propagation combines temporal traces with neuron-specific signals [18]. One number shared across neurons carries less task information than separately assigned neuronal signals, although each synapse still has its own eligibility [19–21]. Recent causal evidence for distinct instructive signals across neuronal populations motivates asking what additional distinctions are useful within a neuron’s arbor [22].

We describe this organization using a *route dictionary*: a set of spatial profiles for delivering learning signals. A route identifies the locations reached by a signal; its profile specifies their relative amplitudes, and a coefficient scales that profile on a given trial. Each dictionary column stores one profile. A shared profile broadcasts throughout an arbor, while several profiles can distinguish subtree addresses. Their linear combinations give the available fields. We use *basis* for linearly independent profiles and *dictionary* when profiles may be redundant. Morphology constrains the available routes; the task and teaching mechanism determine useful coefficients.

Conductance changes the gains of anatomical delivery routes. Excitatory and inhibitory contacts open nonnegative conductances whose reversal potentials determine current; inhibition can lower input resistance and attenuate other

inputs even when its own current is small [23–27]. Rescaling a delivery channel preserves its spatial profile, whereas altered cable transport can reshape how local voltage affects somatic output and task error.

Dendritic models already connect nonlinear compartments and structural plasticity to memory capacity [28], and branch-specific plasticity can organize nonlinear feature combinations [29]. Successive computations in artificial trees and detailed cortical models also establish substantial forward capacity [30, 31]. Dendritic learning theories supply mechanisms for somatic prediction, segregated teaching, interneuron-mediated errors, bursts and prospective dynamics [32–38]; cable-learning models optimize reconstructed neurons using synaptic sensitivities [39]. Our question concerns the spatial information these learning mechanisms must preserve across an arbor.

The conductance-tree gradient factorization and LocalCA rule were introduced in our predecessor preprint [40]. This Article extends those foundations with controlled task-to-credit comparisons, matched-spectrum and conductance learning experiments, and anatomical transport and response tests.

We compare the value of preserving neuronal identity, branch selection, ancestry grouping and example-dependent credit across interacting branches. Morphology supplies candidate patterns and conductance regulates their expression. We establish forward compatibility before interpreting learning gaps; measured responses test the anatomical proposal. The question is which distinctions useful credit must preserve, rather than whether finer feedback is always better.

Results

Exact gradients separate local sensitivity from downstream credit

A useful synaptic update combines local sensitivity with the effect that the change would have on task error. For a point neuron $y_u = f_u(\sum_i w_i x_i + b_u)$, the chain rule gives $\partial \mathcal{L} / \partial w_i = x_i f'_u \delta_u$, where x_i and w_i are synaptic input and weight, b_u is bias, and $\delta_u = \partial \mathcal{L} / \partial y_u$ is the neuronal error. The derivative f'_u is evaluated at the current input sum. Thus local eligibility $x_i f'_u$ and task-derived credit supply different information (Fig. 1A).

We replace the point neuron by a rooted tree with spatially distinct addresses, with soma $n = 0$, compartment voltage V_n , and nonnegative synaptic conductances g_i . A contact with reversal potential E_i^{rev} and nonnegative activity x_i contributes current $g_i x_i (E_i^{\text{rev}} - V_n)$. In the directed artificial trees, child outputs drive their parents and a terminal-to-soma pass computes the steady state. Total input resistance R_n^{tot} is the reciprocal of the summed leak, excitatory, inhibitory and child-coupling conductances (Methods). Differentiating the current balance yields

$$\frac{\partial \mathcal{L}}{\partial g_i} = \underbrace{x_i R_n^{\text{tot}} (E_i^{\text{rev}} - V_n)}_{\text{local eligibility}} \underbrace{\delta_{n,u}^V}_{\partial \mathcal{L} / \partial V_n}. \quad (1)$$

Both excitatory (E) and inhibitory (I) contacts obey this factorization (Fig. 1A). Eligibility tells how conductance changes local voltage; compartment credit tells how that voltage changes task loss.

Here voltage error is a loss derivative, not a voltage difference. In a directed tree, each compartment has one path to the soma. Its exact voltage error is

$$\delta_{n,u}^V = \delta_{0,u}^V \tilde{\alpha}_n, \quad \tilde{\alpha}_n = \prod_{(i \rightarrow k) \in \text{path}(n \rightarrow 0)} f'_i(V_i) R_k^{\text{tot}} g_{i \rightarrow k}^{\text{den}}. \quad (2)$$

Here $f'_i(V_i)$ is the child compartment’s output derivative, $g_{i \rightarrow k}^{\text{den}}$ is its coupling conductance to parent k , and $\delta_{0,u}^V = f'_0(V_0) \delta_u$ is the somatic voltage error. Thus the path multiplier $\tilde{\alpha}_n$ converts one somatic error into a spatial field. The conductance-only component of this product is attenuated by additional conductance on a path’s ancestors; branch nonlinearities and forward voltages can change as well (Supplementary Section S1).

Reciprocal cables instead require a joint adjoint calculation of how local currents affect task loss (Methods). These mathematical sensitivities do not establish a biological error-transmission mechanism.

Our local credit assignment (LocalCA) rule retains the local sensitivity but supplies an approximate learning signal to each compartment. A hat denotes this delivered field:

$$\widehat{\nabla}_{g_i} \mathcal{L} = x_i R_n^{\text{tot}} (E_i^{\text{rev}} - V_n) \widehat{\delta}_{n,u}^V, \quad (3)$$

$$\widehat{\delta}_u^V = A_u \mathbf{c}_u. \quad (4)$$

Columns of $A_u \in \mathbb{R}^{N_u \times K}$ are fixed spatial delivery profiles over N_u routed nonsomatic compartments; $\mathbf{c}_u$ contains their example-dependent coefficients. Broadcast, subtree and compartment profiles give increasingly detailed spatial addresses (Fig. 1B). This linear delivery model does not follow automatically from the forward cable equations and does not specify how a circuit estimates the coefficients.

At $K = 1$, the all-ones column $A_u = \mathbf{1}$ broadcasts one coefficient through the arbor; at $K = N_u$, the identity matrix $A_u = I$ can reproduce the full nonsomatic field. Nested subtrees provide intermediate dictionaries. Profile count, rank (the number of independent profiles) and coefficient information are distinct resources. Exact path transport combines one somatic scalar with state-dependent local derivatives; it does not require an independent external error for every compartment.

Capture is the fraction of squared learning-signal magnitude that a dictionary can reproduce. For a nonzero field δ ,

$$\mathcal{C}(A; \delta) = \frac{\|P_A \delta\|^2}{\|\delta\|^2}, \quad P_A = AA^\dagger.$$

Here P_A gives the best least-squares reconstruction from the profiles; $A^\dagger$ is the Moore–Penrose pseudoinverse and $\|\cdot\|$ the Euclidean norm. Thus capture is a fraction of squared field magnitude, not a fraction of correct decisions. The mean of these per-field ratios weights fields equally. In contrast, $\text{tr}(P_A C) / \text{tr}(C)$, with $C = \mathbb{E}[\delta \delta^\top]$, is the fraction of total ensemble energy captured, weighting fields by their energy. Here C is the uncentered second-moment matrix and tr denotes its trace, the sum of diagonal entries. The two capture measures need not agree. Projection chooses the best coefficients in the specified metric; an implemented learning rule need not supply them.

The expansion applies in a specified coordinate. In the nonlinear image models, we route errors of compartment outputs and retain the local output derivative in eligibility; voltage-error capture is reported separately. Identity-output conductance trees use voltage errors directly. Supplementary Table S14 compares the coordinates, coefficient sources and state dependence. Dictionary capacity answers what fields can be delivered. Learning experiments determine which distinctions improve performance. A one-step smoothness bound describes how restricting noisy updates can improve expected progress, but its initialization utility lost to a fixed maximum-budget tree in all twenty seed averages of a 12,800-fit prospective test, in both selection arms. This local utility therefore did not identify the best final morphology (Supplementary Figs. S35 and S46; Sections S3–S4).

One signal per neuron captures most of the image-task learning benefit

We first tested credit resolution on image classification. All image benchmark networks used one dendritic layer and a trainable linear readout. In the primary comparisons, the readout supplied exact neuronal output errors before the feedback intervention. The MNIST comparison used 128 directed $[3, 3]$ trees, each with three proximal and nine terminal compartments, receiving flattened images. Shunting and raw-additive architectures shared the graph, initial contact construction, nonlinearities and training protocol, with architecture-specific initialization. The additive control removed reversal-potential driving forces and the conductance denominator, retaining signed current summation without shunting.

Six rules separated neuronal identity, spatial resolution and learning in the dendritic core (Fig. 1B,C). A strict scalar shared one signal across neurons; neuron-specific feedback broadcast each neuron’s own output error through its arbor. A projected one-profile rule averaged the exact activation errors over all twelve nonsomatic compartments, while three subtree profiles averaged them within each proximal compartment and its three children. Both projections acquired coefficients from the same exact field and preserved its common-mode mean at identical parameters, although their field norms could differ. Exact transport retained every compartment error. A decoder-only reference trained the readout while freezing the core. Three development seeds selected each rule’s rate from three common

multipliers; ten fresh paired seeds per architecture evaluated selected and original common rates over 180 epochs (Methods; Supplementary Fig. S49A–D).

At selected rates, neuron-specific feedback exceeded the strict scalar by 9.23 percentage points in shunting trees and 7.46 points in additive trees, reaching 97.12% and 97.11% accuracy. Decoder-only learning reached 80.96% and 89.57%, respectively (Fig. 1C). Increasing the projected dictionary from one to three profiles changed accuracy by only 0.049 points in shunting trees (95% paired interval, -0.048 to 0.141) and 0.036 points in additive trees (-0.056 to 0.122 ; Fig. 1D). The common-rate comparison gave similarly small differences (Supplementary Fig. S49E). Exact transport added 0.010 points over neuron-specific feedback in shunting trees, but 0.182 points in additive trees (0.114 – 0.252 ; positive in all ten seeds). Both exact transport and one projected profile reached about 97.3% in additive trees. Most learning benefit therefore came from preserving neuronal coordinates; the smaller additive improvement was attainable without separating subtrees.

The original fifteen-seed cohort and exact-gradient checks remain in Supplementary Fig. S45. Its trained fields retained spatial structure: moving from one to three profiles raised mean voltage-credit capture from 0.480 to 0.625 in shunting trees and from 0.519 to 0.718 in additive trees. The new checkpoint analysis likewise found improved capture without a corresponding accuracy gain from subtree distinctions (Supplementary Fig. S49F). It reports activation-space capture, matching the trained projections, with voltage-space results supplied separately. These reconstructions measure available capacity, not the quality of a biological coefficient encoder. Separate recipient-permutation controls also reduced accuracy while preserving signal values and bandwidth, implicating neuronal identity (Supplementary Fig. S45F,G).

With fixed random soma-level feedback, neuron-specific signals improved over scalar feedback by about eleven points, while exact paths added at most 0.09 points (fifteen paired seeds; Fig. 1E; Supplementary Fig. S30). This control changes weight transport but retains the single dendritic layer and the same output-error dimensionality. Fashion-MNIST also favored neuronal feedback. On flattened CIFAR-10, exact paths reduced accuracy relative to neuronal feedback by 0.86 points (0.55 – 1.17), while remaining equivalent to matched backpropagation within the predefined margin (Fig. 1E; Supplementary Fig. S4).

These benchmarks show that a spatially structured error field need not confer a learning advantage on finer delivery. Supplementary Figs. S4 and S8 retain further depth and task-generator controls. We next introduce task structure that makes particular spatial distinctions useful.

Branch-selective information prevents updates from inactive distractors

To isolate branch-selective information, we used one logistic output with $B \in \{2, 4, 8\}$ parallel branch weight vectors and no internal forward compartments or conductance nonlinearity (Fig. 2A). Context selected the branch determining the output, while every branch received an image and formed an ungated local eligibility.

Inputs were average-pooled, standardized Fashion-MNIST images from classes 0 and 6. The selected image determined the binary target. At conflict dose $\chi = 0$, all branches received that image; increasing χ independently replaced nonselected inputs with opposite-class images. For example i , let b_i^* be the selected branch, $\mathbf{x}_{i,b}$ its input vector and $y_i \in \{0, 1\}$ the target. The branch weights $\mathbf{w}_b$ give the pre-probability score (logit) and output error

$$z_i = \mathbf{w}_{b_i^*}^T \mathbf{x}_{i,b_i^*}, \quad \delta_i = \sigma(z_i) - y_i. \quad (5)$$

Here $\sigma(z) = 1/(1 + e^{-z})$ is the logistic function. The exact derivative of every unselected branch is zero, irrespective of its input. Increasing conflict therefore does not create opposing nonzero exact branch gradients. It makes a rule that ignores the selector increasingly sensitive to irrelevant, opposite-class eligibilities.

For N examples, correctly selected and neuron-shared gradient estimates were

$$\mathbf{d}_b^{\text{selected}} = N^{-1} \sum_i \delta_i \mathbf{1}[b_i^* = b] \mathbf{x}_{i,b}, \quad \mathbf{d}_b^{\text{shared}} = N^{-1} \sum_i \delta_i B^{-1} \mathbf{x}_{i,b}. \quad (6)$$

The indicator $\mathbf{1}[\cdot]$ is one when its condition holds, zero otherwise; $1/B$ matches scale without supplying branch information. A feedback selector and a branch-local eligibility gate give the same update (Fig. 2B), testing access to selection rather than a need for independently computed external errors.

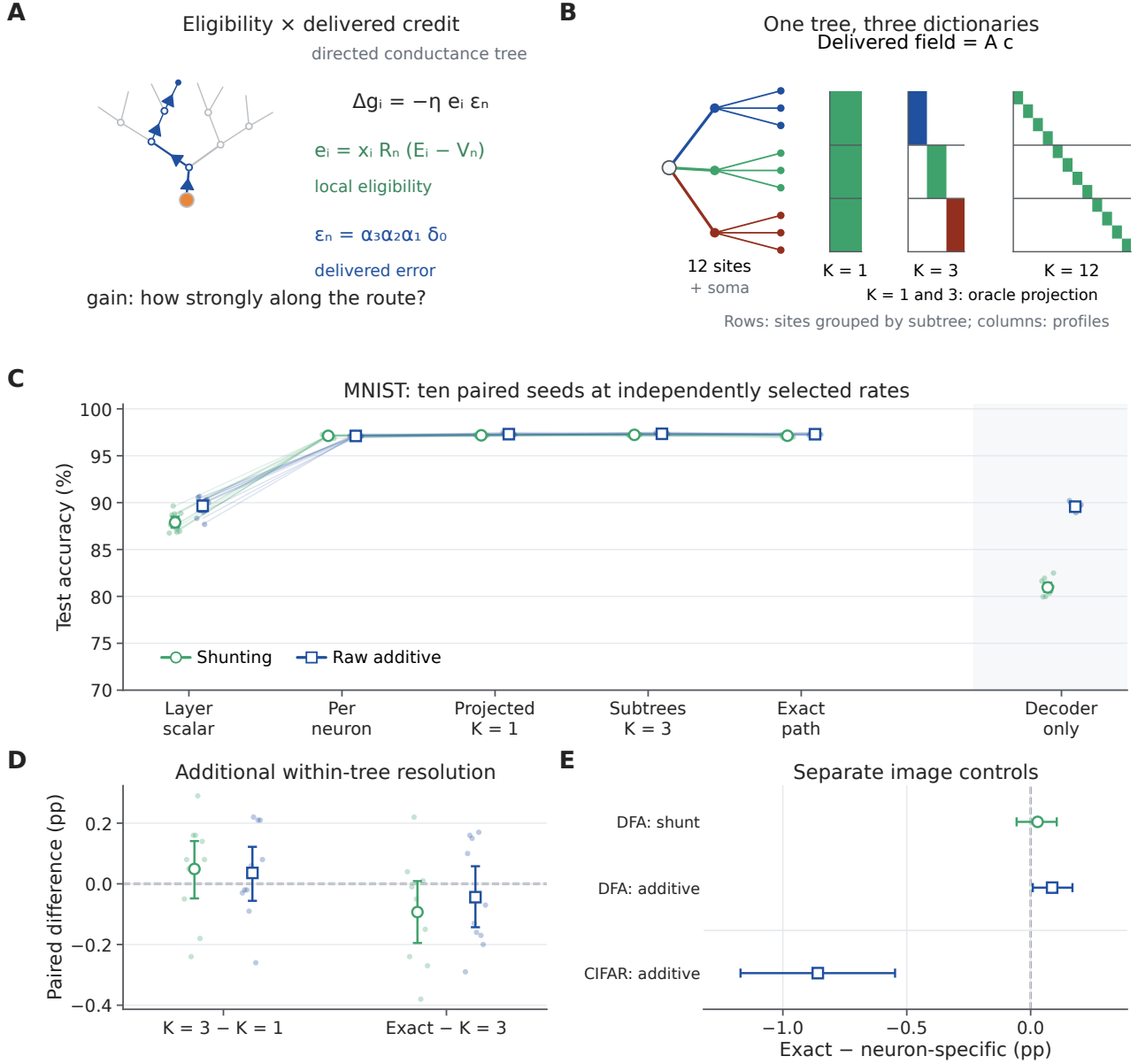


Figure 1: Task-derived credit separates neuronal identity from resolution within a dendritic tree. **A**, Conductance updates combine local eligibility e_i and delivered compartment credit ϵ_n . Here η is learning rate, x_i presynaptic activity, R_n input resistance, and $E_i - V_n$ driving force. Exact directed-tree transport multiplies somatic error δ_0 by path derivatives α ; reciprocal cables require an adjoint solution. **B**, Over the twelve nonsomatic sites, $K = 1$ broadcasts, $K = 3$ groups each proximal site with its three children, and $K = 12$ resolves every site. Projected rules retain exact somatic errors. The $K = 1$ and $K = 3$ training conditions use oracle projection of the exact activation-error field. K counts spatial profiles, not independently computed external errors. **C**, MNIST with one 128-neuron dendritic layer and a linear readout. Neuron-specific credit broadcasts each neuron's exact somatic activation error; the strict scalar removes neuronal identity. The decoder-only reference freezes dendritic parameters. Thin lines pair ten fresh seeds per architecture; large symbols show means and 95% descriptive intervals. All fits use 180 epochs, rates selected on three separate development seeds, and validation-selected checkpoints. **D**, Paired accuracy differences compare $K = 3$ with projected $K = 1$, and exact paths with $K = 3$. The projected rules use the same exact-field coefficient source and preserve the same mean at identical states, but their norms can differ. Points show seeds with paired 95% intervals. **E**, Separate legacy cohorts compare exact paths with neuron-specific broadcast under fixed random between-neuron feedback on MNIST (direct feedback alignment, DFA; 15 seeds) or a readout derivative on flattened CIFAR-10 (20 seeds). Colors and shapes follow **C**. Percentage points are abbreviated pp; original 15-seed readout-derived MNIST ladders remain in Supplementary Fig. S45. Supplementary Fig. S49 adds common-rate and capture controls.

158 A simplified calculation asks when the selected branch’s evidence is outweighed by conflicting evidence
 159 sent to other branches. With centered, symmetric class evidence, the shared-mode prediction crosses zero at
 160 $\chi_c = B/[2(B - 1)]$: 1, 2/3 and 4/7 for two, four and eight branches (Fig. 2C; Methods). More branches therefore
 161 predict an earlier loss of compatibility. This average calculation is distinct from the conflict dose at which a trained
 162 model reaches chance accuracy.

163 At zero conflict, selected and shared rules achieved similar accuracy. Their difference increased with conflict,
 164 reaching 35–58 percentage points at full conflict across the tested branch counts (20 paired seeds; Fig. 2D). In every
 165 seed, the advantage of correct routing increased with conflict. A cyclic reassignment sent every signal to the wrong
 166 branch at matched rank and sparsity. It was harmful even at zero conflict, reducing accuracy by 12–25 points relative
 167 to correct selection. Analytic backpropagation and an explicitly gated grouped-point model matched correct selection
 168 exactly, as required by their algebraic equivalence (Fig. 2E; Supplementary Fig. S29).

169 The shared rule reached chance at lower conflict as branch count increased, consistent with the predicted shift
 170 (Fig. 2F); five two-branch seeds never crossed in the sampled range. Theoretical, mean-curve and individual-seed
 171 crossings are distinct. Gradient alignment also changed during learning (Methods; Supplementary Fig. S29D–F).
 172 Credit reaching inactive branches can therefore harm learning; dendrites offer one possible selection site. A separate
 173 two-stream task tests learning and interference effects (Supplementary Fig. S19).

174 **Ancestry provides a modest advantage at an intermediate feedback budget**

175 Branch selection compares fully shared with fully resolved feedback. We next asked whether nested subtree routes
 176 offer a useful intermediate dictionary. The task contained eight independent, eight-dimensional terminal weight
 177 blocks at the leaves of a balanced binary feedback tree. Only the context-selected block determined the logit. Internal
 178 tree nodes defined feedback supports and did not perform serial forward computation (Fig. 3A).

179 The selected stream carried the class signal. Distractors carried opposing signals that strengthened with distance
 180 through the feedback hierarchy, deliberately favoring task-aligned grouping. At each budget $K \in \{1, 2, 4, 8\}$, the
 181 routing field had K independent patterns and unit-length rows. Ancestry routes grouped the terminals into the
 182 whole tree, halves, sibling pairs or individual terminals. Permuting task streams across leaves changed alignment
 183 without changing the streams or forward parameter budget. Here bandwidth counts independent feedback patterns,
 184 not physical stages or independent scalar losses; other single-pattern controls need not be uniform. The complete
 185 generator and update are given in Methods.

186 The generator predicts where an intermediate ancestry benefit can occur. The selected stream has class coefficient
 187 one; its sibling, same-half nonsiblings and other-half streams have coefficients -0.15 , -0.45 and -0.75 . The
 188 unnormalized sum of the class signal within the selected ancestry group gives -3.05 , -0.05 , 0.85 and 1 at
 189 $K = 1, 2, 4, 8$, respectively. Thus coarse pooling reverses or nearly cancels the signal, sibling-pair routing first
 190 preserves a positive intermediate-group signal, and full resolution removes the grouping distinction. The implemented
 191 unit-row normalization divides these sums by the square root of group size, preserving their signs. This is an analytic
 192 consequence of the fixed task design; its location is not an independently discovered optimum (Fig. 3B).

193 Controls preserved rank while changing recipient assignment, support geometry or sparsity: cyclic derangement,
 194 depth-interleaving, random-sparse and dense random fields. The per-seed best-control summary takes their maximum
 195 after training; it is not a learned selector. A training-informed principal-component reference is retained separately
 196 (Methods).

197 Increasing bandwidth improved correct routing, while derangement caused large losses (Fig. 3C). Degree- and
 198 depth-preserving rewiring removed the task-aligned advantage at intermediate budgets but did not affect fully shared
 199 or fully resolved ancestry routes (Fig. 3D).

200 The ancestry-specific effect was much smaller. Ancestry performed worse than the best matched control at
 201 low budgets, exceeded it by 1.27 percentage points at $K = 4$, and tied at full resolution. The paired $K = 4$ interval
 202 was 0.59–1.95 points, with 15/20 positive differences; a paired Wilcoxon signed-rank test gave Holm-adjusted
 203 $P = 0.0101$ across four budgets. Individual control contrasts and paired seed differences distinguish this effect from
 204 the much larger correct-versus-deranged contrast (Fig. 3E,F).

205 Flat, grouped-point and gated-point representations receiving the same field behaved identically by construction.

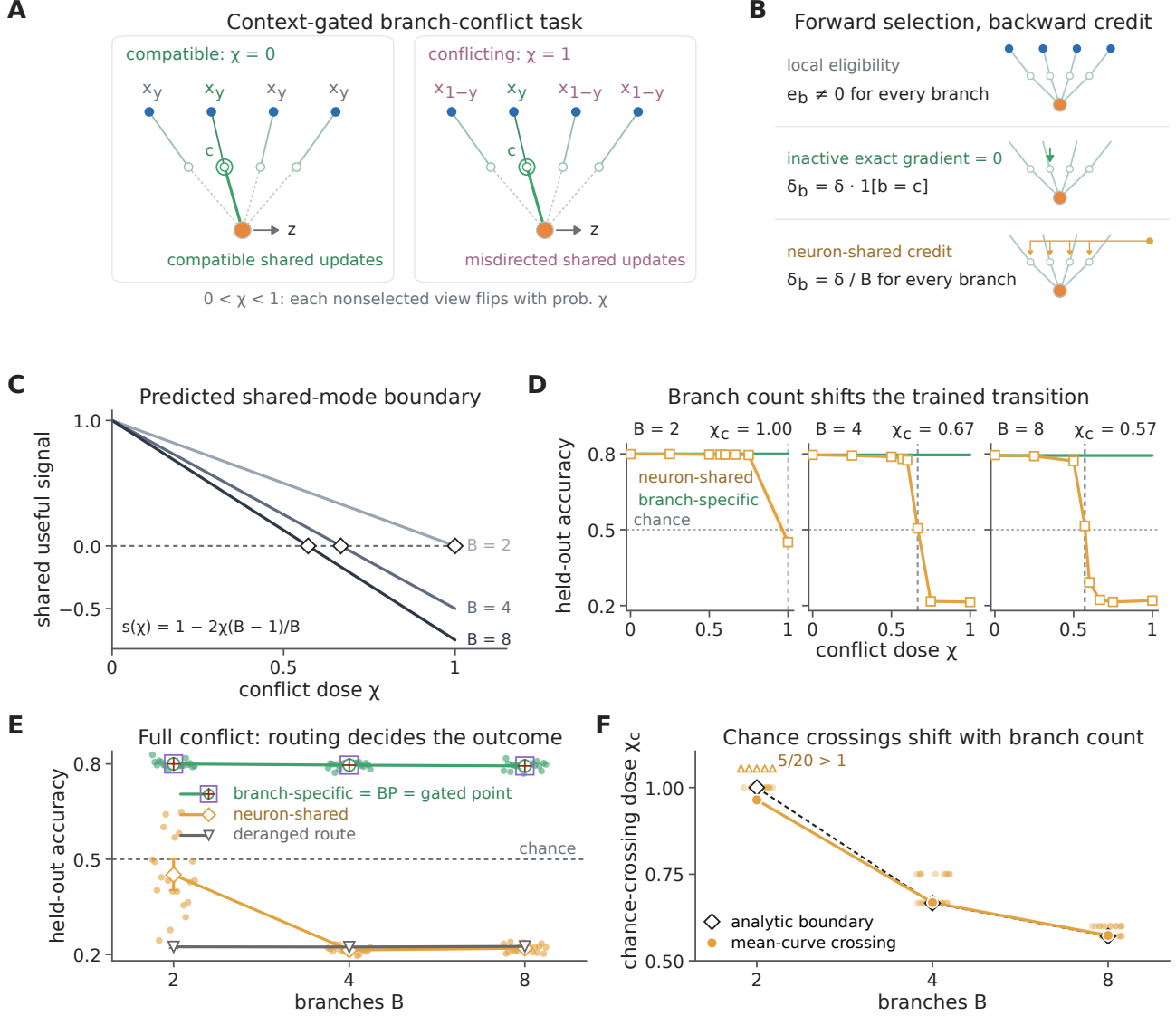


Figure 2: Branch selection prevents learning from inactive distractors. One logistic output combines B parallel branch weight vectors; there are no serial or conductance nonlinearities. **A**, Context-gated Fashion-MNIST task. Context c selects the branch determining logit z ; every branch receives an image and forms an ungated eligibility. Conflict dose χ independently replaces each nonselected image by an opposite-class image. Solid and dashed proximal paths are selected and gated-off forward routes; the double ring marks context. **B**, Selected feedback: $\delta_i \mathbf{1}[b = b_i^*]$, shared feedback: δ_i/B . Here δ_i is the output error, b_i^* the selected branch and $\mathbf{1}$ the indicator; unselected exact derivatives are zero. A local eligibility gate gives the same update. **C**, Centered symmetric-evidence prediction $s(\chi) = 1 - 2\chi(B - 1)/B$: shared-mode compatibility vanishes at $\chi_c = B/[2(B - 1)]$. This mean-field boundary is distinct from a trained chance crossing. **D**, Held-out accuracy over conflict dose. **E**, At full conflict, correct selection, backpropagation (BP) and the algebraically equivalent gated-point calculation coincide; cyclic derangement sends signals to the wrong branch at matched rank and sparsity. Selection-by-conflict interactions are positive in all 20 seeds for each B . **F**, Analytic threshold (diamonds), first interpolated crossing of mean accuracy (orange), and each seed's first sampled at-or-below-chance dose (faint dots). Open overflow triangles retain five $B = 2$ seeds that do not cross within the sweep. Curves and bars summarize 20 paired seeds with 95% seed-bootstrap intervals. Branch-count comparisons do not hold capacity fixed.

206 The experiment therefore supports a conditional advantage of ancestry-shaped feedback under limited bandwidth,
 207 without attributing the effect to dendritic material or treating equivalent representations as independent confirmations.
 208 Learning the route coefficients was another limitation. In twenty fresh seeds, an estimator trained on separate
 209 noisy context cues improved accuracy over a frozen profile by 7.16 points, but remained 54.47 points below oracle
 210 context coefficients. Exploratory selection of its most probable route narrowed the gap, whereas delaying cues
 211 removed the benefit (Supplementary Fig. S32; Methods). These tests expose requirements for selective, timely
 212 delivery; they do not derive the context signal from local eligibility.

213 **Cross-branch interactions change the spatial credit that learning uses**

214 Interactions across branches can change credit’s sign and magnitude between examples. We compared targets with
 215 identical input-sensitivity spectra on the same representationally compatible tree, separating credit delivery from
 216 forward capacity.

217 Eight independent inputs x_i were equally likely to be -1 or $+1$. Each branch computed $u = a + bu_L + cu_R + du_Lu_R$
 218 with four trainable coefficients. This multi-affine unit is affine in either child while the other is fixed. Pairwise
 219 targets summed four disjoint products, $f_M = \frac{1}{2} \sum_{(i,j) \in M} s_{ij} x_i x_j$; quartic targets combined complementary quartets,
 220 $f_A = \frac{1}{2} (s_A \prod_{i \in A} x_i + s_{\bar{A}} \prod_{i \notin A} x_i)$. Here M partitions inputs into pairs, A is one quartet, $\bar{A}$ its complement, and signs s
 221 were independently sampled. Within each seed, targets shared a compatible balanced tree, input assignment, initial
 222 weights and examples (Fig. 4A). Both have input-sensitivity second moment $\mathbb{E}[\nabla_x f \nabla_x f^T] = I_8/4$, giving identical
 223 strengths of independent sensitivity directions.

224 Compatibility has an exact criterion. At each subtree boundary, arrange interaction coefficients into a matrix
 225 indexed by products of inputs inside and outside the subtree. Removing the constant inside row must leave rank at
 226 most one, because only one scalar leaves the branch. This condition is necessary and sufficient across all boundaries;
 227 discarded singular values bound representation error (Methods; Supplementary Section S4). It constrains forward
 228 capacity, not the learning rule. Boolean examples and the structure census appear in Supplementary Figs. S36–S37
 229 and S43–S44.

230 All credit rules used the same exact local eligibilities and scalar output error. We compared exact paths with
 231 unit broadcast and a fixed six-site profile calibrated from each student’s mean path derivatives over 256 unlabeled
 232 initialization examples. Three development seeds selected learning rates with equal budgets; twenty fresh paired
 233 seeds tested the frozen choices. A separate nested target summed products over prefixes of two, four, six and eight
 234 inputs on its own compatible tree.

235 After 1,024 Adam steps, exact and calibrated credit gave mean held-out normalized mean squared errors (NMSE)
 236 of 0.0317 and 0.0236 for pairwise targets, 0.1516 and 0.9813 for quartic targets, and 0.0230 and 0.7690 for nested
 237 targets (Fig. 4B–D). NMSE divides squared prediction error by clean target variance. Exact credit had lower
 238 pairwise error in 15 of 20 seeds; one slow fit reversed the mean ordering. Early curves also compare different
 239 development-selected rates: 0.01 for the profile and 0.003 for exact credit; the profile does not retain an early
 240 advantage at every common-rate checkpoint (Supplementary Fig. S54A,B). The predefined quartic-minus-pairwise
 241 difference in calibrated-minus-exact gaps was 0.838 (95% paired seed-bootstrap interval, 0.719–0.931), positive in
 242 all twenty seeds; a common rate gave 0.843 (0.727–0.935; Fig. 4E). Exact quartic outcomes separated into seventeen
 243 fits near the noise floor and three stalled fits, so their mean alone obscures substantial variation. Equal label noise
 244 gave different NMSE floors because pairwise and quartic target variances were one and one-half.

245 A balanced extension replayed all 720 algebraic trajectories and continued them to 16,384 updates without
 246 retuning rates. All twenty selected-rate Adam exact quartic fits reached the noise floor by 4,096 updates, and the
 247 pairwise mean lag disappeared. At the longest budget, the quartic-minus-pairwise calibrated-credit deficit was 0.952
 248 (95% paired interval, 0.883–1.012) at terminal states and 0.790 (0.691–0.880) under validation-based selection,
 249 positive in all twenty seeds. In these selected-rate Adam comparisons, fixed-profile arms contacted coefficient
 250 bounds, whereas exact arms did not. These are descriptive extensions of previously observed seeds under the same
 251 bounded dynamics (Supplementary Figs. S54–S55 and Table S37).

252 The trained fields reveal a distinction missed by the input spectrum. Let $\mathbf{q}(x)$ contain the six nonsomatic
 253 derivatives of root output with respect to branch output; exact branch credit is scalar output error times $\mathbf{q}$. Allowing

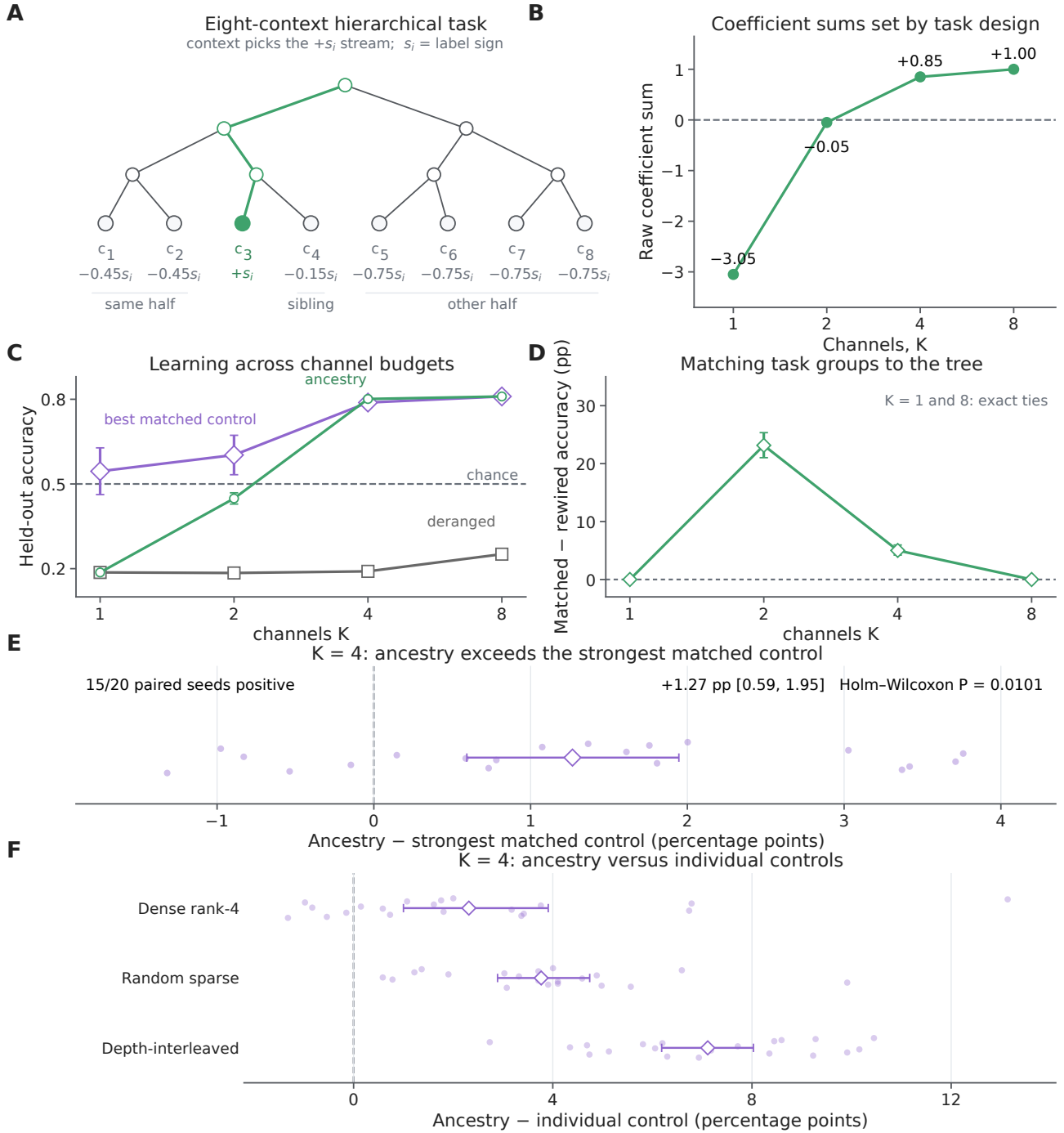


Figure 3: Hierarchical distractors favor ancestry at intermediate bandwidth. **A**, Eight streams carry label sign s_i along noisy teacher features. The selected stream has coefficient one; sibling, same-half and opposite-half distractors have coefficients -0.15 , -0.45 , -0.75 . **B**, Coefficient sums in the cued ancestry group at $K = 1, 2, 4, 8$. Implemented feedback divides these sums by square root of group size, preserving their signs. The four-route sign change follows from the design; it is not an independently discovered accuracy optimum. **C**, Held-out accuracy for correct ancestry, derangement and the per-seed maximum over four matched controls. This maximum is calculated after training, not learned by a selector. Dashed line: chance. **D**, Task-matched-minus-rewired accuracy at equal degree and depth. **E**, The four-route ancestry-minus-best-control effect, 1.27 pp (95% interval, 0.59–1.95), with 15/20 positive seeds and Holm-adjusted two-sided Wilcoxon $P = 0.0101$ across four budgets. **F**, Individual dense, sparse and depth-interleaved contrasts on a separate scale. The large derangement effect is shown in **C**. Symbols and bars are means and 95% seed-bootstrap intervals; small points in **E,F** are twenty paired seeds. Endpoint ties follow from coincident operators.

an oracle amplitude per example, the best fixed profile captured 99.7% of exact-trained path-field energy for pairwise targets, but 46.1% for quartic and 44.8% for nested targets (Fig. 4F). Loss-error weighting, eligibility weighting, site-normalized spectra and replay of 600 earlier fits preserved this separation (Supplementary Section S4). Uniform-profile capture did not: a nearly one-dimensional field can have a signed, nonuniform direction. Positive fixed-profile scaling largely cancels in Adam, explaining calibrated/sign-only agreement; it does not explain the slower pairwise learning with changing exact paths.

Thus input-matched targets developed different credit spectra and tolerance of fixed profiles. At fixed weights and inputs, however, path derivatives are target-independent: their spectrum describes learned states and coordinates, not a task-only initialization diagnostic. Failure of the tested profiles supports example-dependent spatial credit without establishing that six independent channels or exact gradients are necessary.

Additional controls delimit the result. Fixed-profile Adam learned a compatible positive-conductance teacher accurately; broadcast learned aligned XOR-of-AND, whereas parity favored exact credit under the tested optimizer (Supplementary Figs. S42–S44). Noisy-query interaction estimation also yielded useful trees, retained as supporting structure-design evidence in Supplementary Figs. S41–S42. Interaction structure therefore motivates specific credit comparisons rather than a universal monotonic resolution law.

Context-dependent branch tuning makes spatial credit useful in conductance trees

We next tested how branch tuning affects credit demand in a conductance tree. A first seven-compartment construction with monotonic branch responses was learned accurately by all tested rules. Under Adam, mean broadcast NMSE remained below 0.0006 after extended training despite a varying path field (Supplementary Fig. S50). We therefore tested a separate construction in which context selected branches with either aligned or opposing feature tuning (Fig. 5A,B).

Four terminal compartments fed two proximal compartments and a soma, using the directed E/I conductance balance (Methods). Both subtrees received the same positive feature pairs, $\exp(z_j)$ and $\exp(-z_j)$, for two independent latent inputs z_j . A binary context increased inhibitory conductance in one proximal compartment. Teacher excitation and inhibition produced increasing feature responses in both subtrees for the aligned task, but increasing responses in one and decreasing responses in the other for the opposed task. Teacher and student shared the circuit, ensuring representability. Students shared the teacher’s nominal proximal and coupling gains, with independent Gaussian perturbations of SD 0.4 in log conductance; terminal gains started near one. All 24 conductances were plastic, and paired conditions shared initial weights and example streams.

At fixed weights, binary context generates at most two exact path profiles. Their gains are positive: context changes the relative weighting of the two distal subtrees, while proximal credit is context invariant. The nominal parent’s conductance denominator changes by approximately eightfold. Thus branch selection is the relevant distinction, connecting this task to Fig. 2.

We tested whether that distinction requires oracle coefficients. A local rule delivered the somatic error to terminal parameters only when their parent was uninhibited, while retaining ungated error at proximal compartments and the soma. It used the supplied inhibitory context and exact local eligibilities, without evaluating exact paths or projecting an exact field. Two fixed distal profiles with oracle amplitudes and unit proximal credit supplied a separate capacity control. An additional proximal profile in the earlier three-profile control therefore addresses proximal updates, rather than another context distinction.

After exploratory panel results motivated this rule, we froze its protocol and tested twenty new paired seeds. All nine rules ran at three common rates through 16,384 updates; the primary rate and 4,096-update window were fixed before these outcomes. On opposed targets, mean held-out NMSE was 2.18×10^{-5} for exact credit, 2.33×10^{-5} for the local gate, 2.28×10^{-5} for the two-profile oracle with unit proximal credit, and 0.487 for broadcast (Fig. 5C,D). On aligned targets, the corresponding exact, local-gate and broadcast errors were 7.53×10^{-6} , 1.06×10^{-5} and 5.53×10^{-4} . The opposed-minus-aligned difference in broadcast-minus-gate error was 0.487 (95% paired interval, 0.360–0.615), positive in all twenty seeds. It remained 0.468 (0.332–0.604) within the longer window (Fig. 5E).

The controls identify what the gate must preserve. Reversing its distal selection gave opposed-task NMSE 0.949. Applying it at proximal compartments as well gave 0.123 and prevented both proximal inhibitory gains from

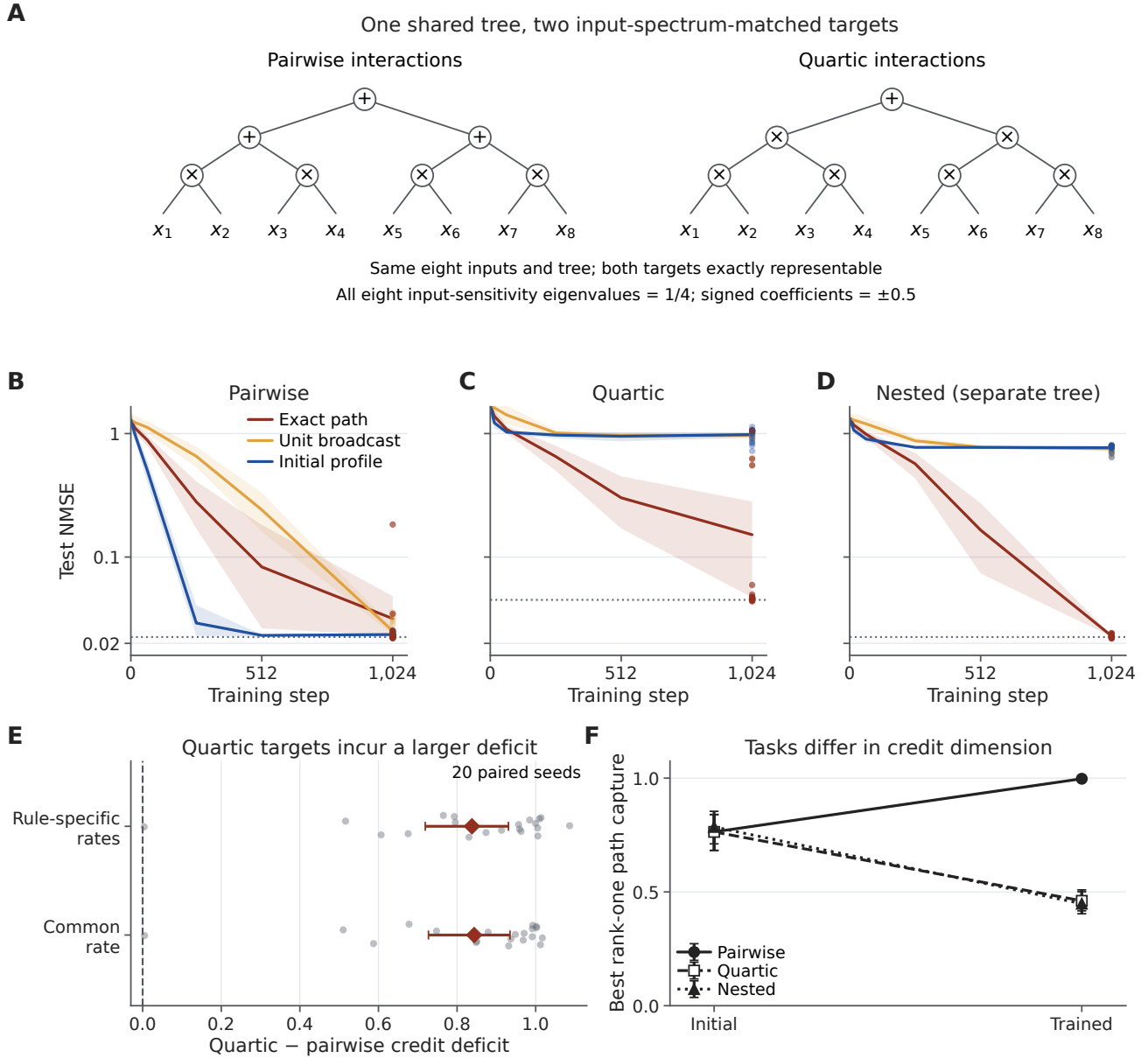


Figure 4: **Input-spectrum-matched tasks differ in learned credit geometry and tolerance of fixed spatial credit.** **A**, Pairwise and quartic targets share eight inputs, a balanced tree, initialization and example streams. Drawings show interaction supports; coefficients have random signs and magnitude 0.5, with input labels permuted together within each seed. Both targets are exactly representable and have input-sensitivity second moment $I_8/4$. **B–D**, Held-out NMSE during Adam training with exact path credit, unit broadcast and an initially calibrated profile. The latter averages the initial path derivatives on 256 unlabeled examples and remains fixed. The somatic derivative is one for all rules. Nested targets use a separate compatible tree. Rates were selected using three development seeds per rule. Lines and bands show means and descriptive 95% whole-seed bootstrap intervals; endpoint dots retain all twenty fresh seeds. Dotted lines indicate noise-only NMSE: 0.0225 for pairwise and nested, 0.045 for quartic targets. **E**, Quartic-minus-pairwise difference in the calibrated-minus-exact NMSE gap, at rule-specific rates and a common rate of 0.003. **F**, Best rank-one capture of six-site path-field energy before and after exact-credit training. The spatial direction and example-specific amplitudes are oracle fits, not an implemented encoder. Points and intervals in **E,F** summarize twenty paired seeds. All fits stop at 1,024 steps.

learning: their eligibility is nonzero only during inhibition, precisely when that gate closes (Fig. 5F). The proper local rule changed all 24 parameters and succeeded across the three tested rates. A continuous shunt-proportional distal gate also worked (Supplementary Fig. S53).

This success is consistent with coarsening the exact rule’s strong context-dependent gain ratio. Earlier same-state gradient diagnostics showed that broadcast mixes opposing context-conditioned updates, despite nonnegative per-example alignment with exact gradients (Supplementary Fig. S53E). The result establishes a local implementation when inhibitory context and somatic error are supplied; it does not explain their biological origin. All comparisons remain finite-budget and bounded-parameter results. Exact and local-gate fits did not reach conductance bounds, whereas most opposed broadcast fits did; the earlier wide-bound and rate controls remain in Supplementary Figs. S51–S52.

Serial computation helps learning, while credit benefits depend on budget and metric

The interaction comparison holds a compatible forward tree fixed while changing credit. We next varied serial forward structure itself, asking whether successive compartments help remove unwanted multiplicative changes in input strength. Physical depth D_p counts forward stages, whereas feedback resolution counts distinctions in the backward field.

Each network contained 64 somata and eight nonsomatic compartments per soma. D1 [8], D2 [2,3] and D3 [2,1,2] distributed those compartments across one to three stages while preserving contacts and trainable resources (Fig. 6A). Branching vectors are ordered from soma to tips. A grouped-point control connected all eight modules directly to the soma; raw-additive trees removed divisive conductance normalization. Flexible point networks provided additional, approximately parameter-matched comparisons.

Distal excitation carried class information multiplied by nuisance gains. Inhibitory sensor blocks reported those gains with alignment α : zero retained the sensor distribution but removed trial-specific information, and one reported the factor exactly. With inhibitory reversal zero, the sensors entered the conductance denominator and could support successive divisions [24, 41]. Separate nested, flat and locally cancellable gain tasks tested the role of factor grouping and input access (Methods; Supplementary Section S5).

The original 180-epoch comparison favored aligned serial D3 models and showed a large exact-path advantage within the LocalCA training recipe (Supplementary Fig. S31). However, all fifty D3 fits across the five credit/optimization conditions reached that cap with validation loss still falling. We therefore repeated those fifty fits and ten exact-BP D1 references with a maximum of 600 epochs, preserving the original seeds, learning rates, parameter groups and patience of thirty epochs. Both optimization recipes used Adam. These were restarts, because the historical checkpoints lacked optimizer state; source-concordance differences are retained in SI. Comparisons between budgets use validation-selected states from these same new trajectories (Fig. 6B–E).

Within those restarts, the LocalCA exact-path-minus-shared-soma accuracy gap changed from +10.86 points at the 180-epoch budget to -1.52 at 600 (95% paired interval, -1.83 to -1.23), with this budget-indexed sign change in all ten seeds. The mean gap remained negative from epoch 315, reached -2.94 points at epoch 486, and moved back towards zero by 600 (Fig. 6C). Exact-path LocalCA retained lower mean cross-entropy over epochs 180–600, ending at 0.254 versus 0.276, but this gap also varied non-monotonically (Fig. 6E). Accuracy counts threshold crossings, whereas cross-entropy also reflects predicted probabilities. Under the BP recipe, the 600-epoch exact-minus-broadcast accuracy gap was 0.48 points (0.40–0.55).

The exact-BP D3-minus-D1 accuracy gain was likewise budget dependent: 30.82 points at 180 epochs and 38.17 at 600. Eight of ten D1 references stopped early under the validation-patience rule; the other two and all fifty D3 fits reached 600. D3 validation losses were still falling, and their selected states lay near the cap. Validation-based selection therefore does not establish convergence or equal optimization maturity across depths. Sensor shuffling, reversed placement, grouped-point and task-family controls delimit the serial computational advantage (Supplementary Figs. S15, S18, S28 and S31); same-seed reruns reproduce the original-budget comparisons (Supplementary Fig. S26). These results support useful serial computation under the stated budgets while leaving the eventual credit ordering unresolved.

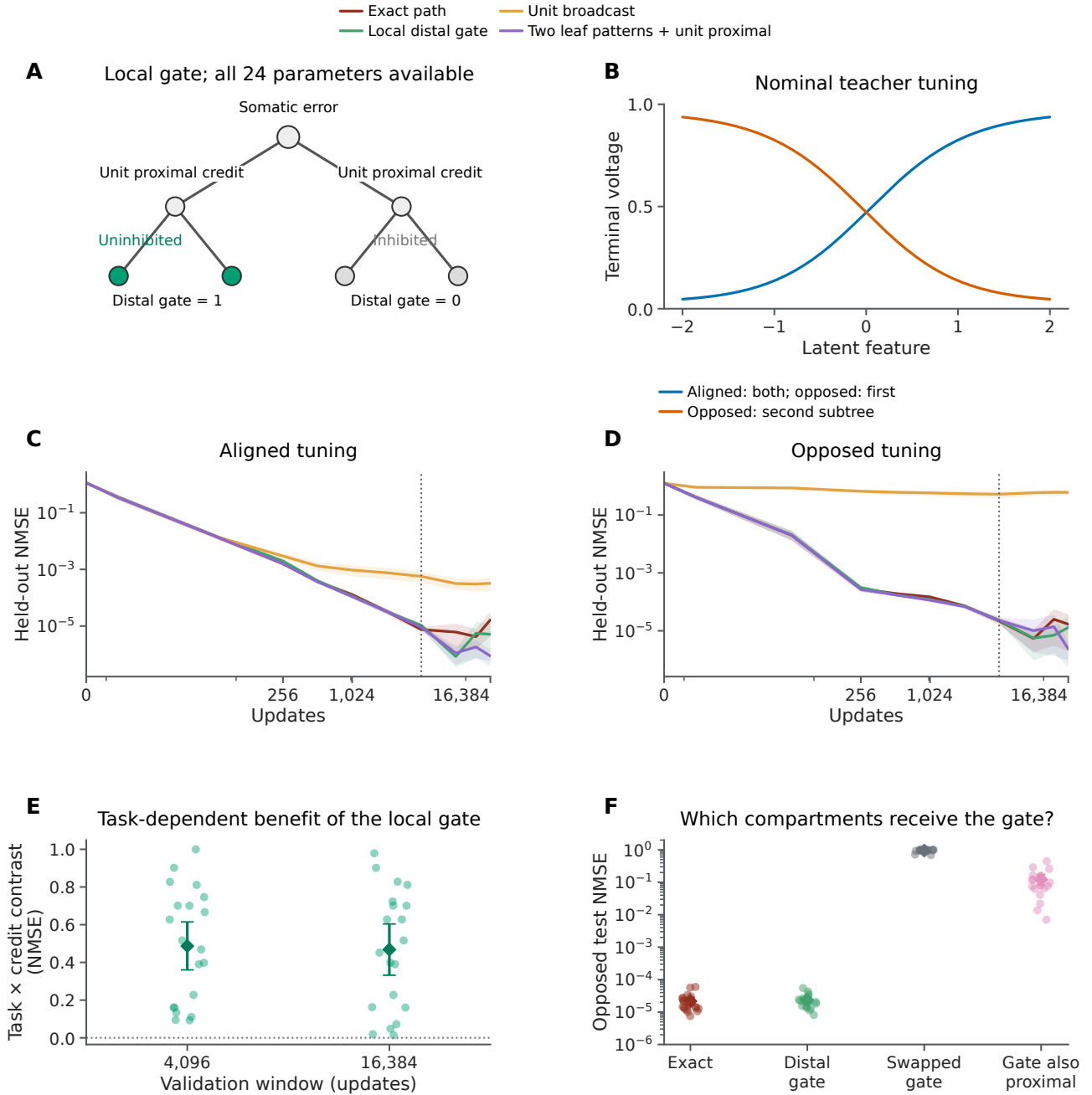


Figure 5: A local inhibitory gate supplies useful branch selection in a conductance tree. **A**, Four terminal compartments feed two proximal compartments and a soma. All 24 conductances are plastic. The local rule gates terminal credit by the parent’s inhibitory activity while retaining unit multipliers for proximal and somatic credit. Local eligibilities are unchanged. **B**, Nominal teacher tuning for aligned and opposed branches; both tasks receive identical positive feature pairs. **C,D**, Fixed-checkpoint test NMSE for aligned and opposed targets, respectively. Twenty new paired seeds share initial students and example streams across rules. Adam rate 0.03 and both budgets were fixed before execution. The two-profile control uses oracle distal amplitudes and unit proximal credit. Dotted lines mark 4,096 updates; every trajectory continues to 16,384. Update axes are linear through 64 and logarithmic thereafter. **E**, Opposed-minus-aligned difference in broadcast-minus-local-gate NMSE, evaluated at minimum-validation-error states within each budget. **F**, Opposed-task outcomes within the primary window for correct, swapped and proximal-gated rules. Gating proximal credit freezes two inhibitory gains. Points show every seed; larger symbols and bands in **C–F** show means and descriptive 95% whole-seed bootstrap intervals. All rates and nine rules are retained in Supplementary Fig. S53. Earlier exploratory gate runs are excluded.

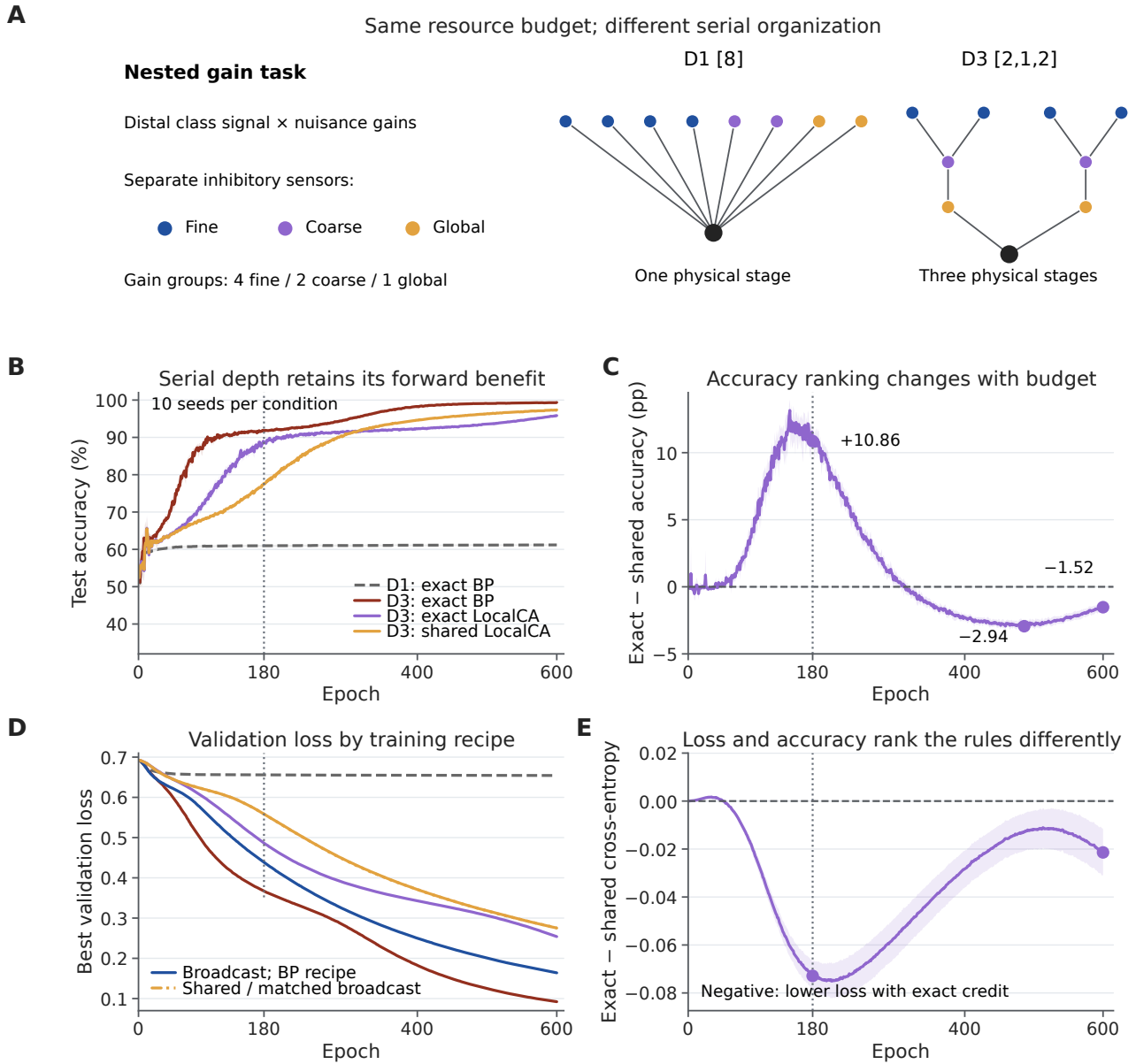


Figure 6: Serial computation remains useful while credit comparisons change with budget and metric. **A**, Nested nuisance gains multiply distal class evidence. Separate inhibitory sensors report four fine, two coarse and one global gain. Colors denote the four/two/two modules assigned these inputs; the drawing indicates input access, not exact cancellation. D1 and D3 distribute eight compartments per soma across one or three stages with matched contacts and resources. **B**, Test accuracy of validation-selected states for D1 exact backpropagation (BP) and three D3 rules. Stopped runs retain their selected state. **C**, Paired D3 exact-path-minus-shared-soma LocalCA accuracy gap, marked at 180, 486 and 600 epochs. **D**, Best validation loss for all six conditions; shared conditions use the colors in **B**. Broadcast autograd with the LocalCA recipe and shared-soma LocalCA nearly coincide and share a legend entry. All fifty D3 fits reach 600 epochs with falling late validation loss; eight of ten D1 references stop earlier. **E**, Paired test cross-entropy gap at the same selected states. Exact LocalCA retains lower mean loss throughout epochs 180–600. Shading in **B,C,E** gives pointwise descriptive 95% whole-seed bootstrap intervals from ten paired seeds. Vertical lines mark 180 epochs. Both recipes use Adam with their original rates and parameter groups; validation selection does not establish convergence.

Reconstructed arbors provide spatial routes beyond a shared broadcast

MICrONS links reconstructed anatomy and connectivity to visual responses [42]. We used observed skeletons, soma positions and mapped contact locations and areas to model conductances, voltages and gradient fields, asking whether measured geometry supplies useful delivery patterns. This tests anatomical capacity rather than observed teaching signals.

Each inhibitory-bearing segment supplied a candidate route over its input-bearing descendants (Fig. 7A). For a site-by-route matrix A_K , diagonal excitatory-area weights W and weighted least-squares projector $P_{A_K, W}$, capture of a field $\mathbf{t}$ was

$$\mathcal{C}(A_K; \mathbf{t}) = \frac{\|P_{A_K, W} \mathbf{t}\|_W^2}{\|\mathbf{t}\|_W^2}, \quad \|\mathbf{v}\|_W^2 = \mathbf{v}^T W \mathbf{v}. \quad (7)$$

For an ensemble of perturbations, numerator and denominator sum over field columns. A common-constrained SVD dictionary provided a ceiling at the same column budget. Controls used random anatomical routes, cable-distance bins, site-shuffled routes preserving nonzero counts, and surrogate trees preserving degree and depth while changing ancestry.

The fields were gradient changes induced by small focal shunts in reciprocal cables, generated without inspecting the selected dictionary. They nevertheless favor ancestry by construction: passive transport under a somatic loss has the ancestry structure derived below. This is a test of how well a restricted dictionary compresses the tree's own perturbation responses. In the original eight cells, broadcast captured 31.4% of field energy, near the 32.6% rank-one ceiling. Because subtree routes originally lacked the constant component available to depth bins, we gave every family one budgeted broadcast and $K - 1$ spatial profiles. The protocol was fixed before extension to 47 disjoint cells from the same mouse and a reconstructed second mouse (Fig. 7).

At eight columns in the 47-cell cohort, broadcast-plus-ancestry captured 0.619 of total energy versus 0.578 for degree–depth surrogates. The paired advantage was 0.041 (95% cell-bootstrap interval, 0.020–0.062), positive in 38 of 47 cells. This cohort mean conceals heterogeneity: in 11 cells, more than half of the 200 surrogates matched or exceeded the actual tree. Depth bins captured 0.511 and site-shuffled profiles 0.436 (Fig. 7B); the ancestry-minus-depth difference was 0.108 (0.077–0.135), positive in 45 cells. After removing broadcast, ancestry and depth-bin residual capture was 0.527 and 0.392 (Fig. 7C). Actual ancestry therefore improves average compression within this passive response family, while coarse geometry explains much of the capacity.

The original eight cells retained the same ordering (Fig. 7D). In the Pinky second mouse, eight of ten qualifying cells supported the eight-column comparison. Their 9–13 input-bearing sites put ancestry near the residual-capture ceiling, so lower-budget comparisons and exclusions remain separate from the larger arbors. Earlier route-generated positive controls test explicitly imposed alignment (Supplementary Figs. S10, S20 and S27).

Direct E/I labels covered 4.71% of disjoint-cohort inputs; the original cohort's 73.0% direct-versus-proxy agreement concerned jointly labeled contacts. Nonzero site-by-route coefficients are a delivery-cost proxy, not feedback synapses or metabolic expenditure (Fig. 7E). Geometry, labeling and compression sensitivities are in Methods and Supplementary Figs. S12, S25 and S33.

Focal shunting sets gains on ancestry-defined credit routes

Morphology supplies spatial routes; conductance can regulate their amplitudes. We examined this connection by comparing a focal inhibitory shunt with a current injection matching its baseline focal current in reconstructed reciprocal cables. A separately solved somatic current restored baseline somatic voltage after either intervention. The endpoint was the change in the modeled excitatory-conductance gradient.

Let G be the passive conductance matrix and $\mathbf{b}$ the current-source vector, so that baseline voltages satisfy $G\mathbf{V} = \mathbf{b}$. With $\mathbf{e}_k$ the unit vector at compartment k , adding inhibitory conductance κ_k at compartment k changes both G and its source: $G' = G + \kappa_k \mathbf{e}_k \mathbf{e}_k^T$, $\mathbf{b}' = \mathbf{b} + \kappa_k E_I \mathbf{e}_k$. The matched injection supplies baseline current $\kappa_k (E_I - V_k)$ without changing G . The adjoint $\mathbf{q}$ gives loss sensitivity to current injected at each compartment. For a fixed somatic loss derivative, it changes by

$$\mathbf{q}' = \mathbf{q} - \frac{\kappa_k q_k}{1 + \kappa_k (G^{-1})_{kk}} G^{-1} \mathbf{e}_k. \quad (8)$$

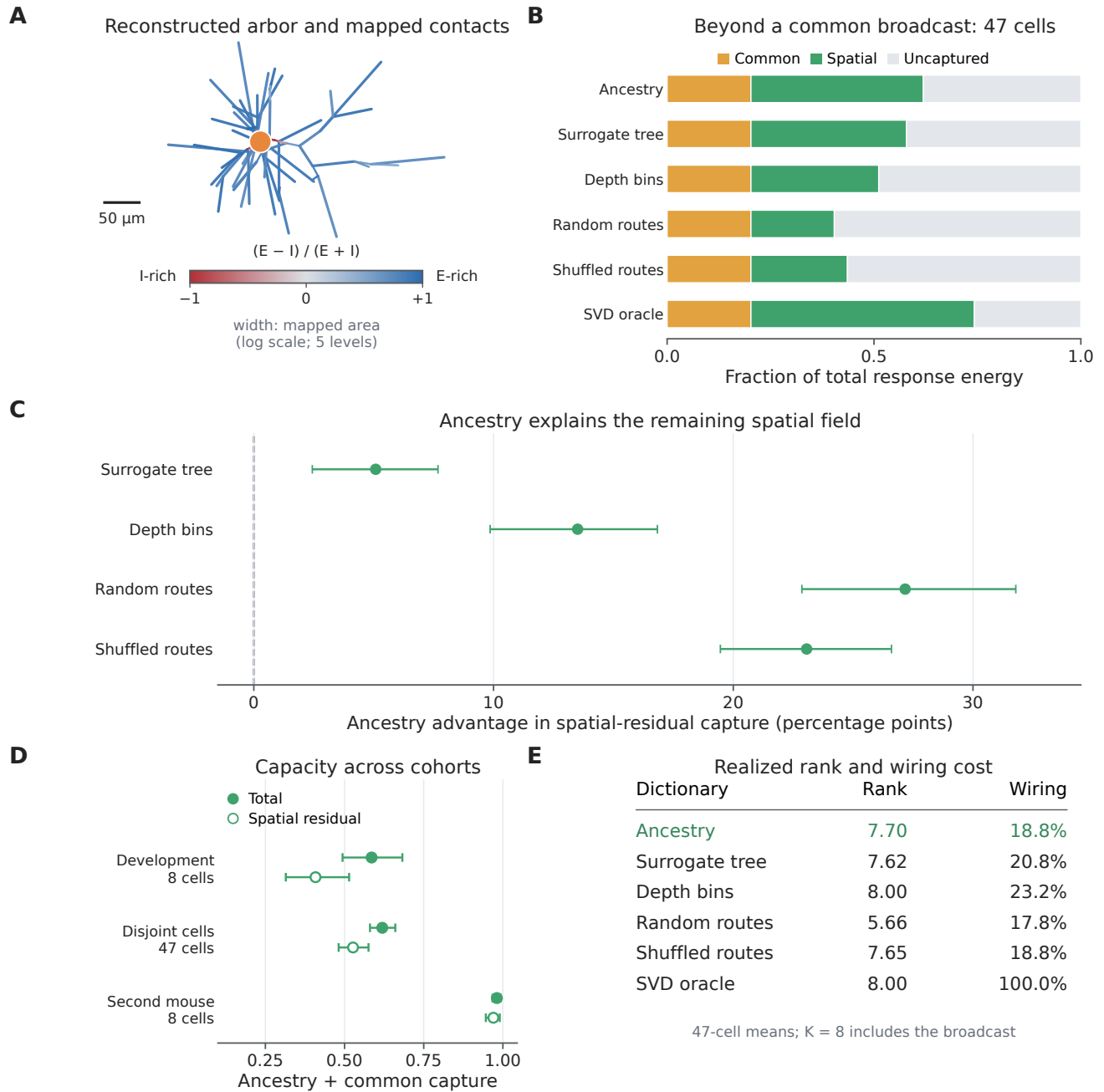


Figure 7: Ancestry captures spatial response structure beyond a shared broadcast. **A**, Median-sized reconstructed arbor from the initial cohort. Color indicates mapped E/I area balance; width indicates total contact area; scale bar, 50 μm . **B**, Mean energy decomposition for modeled focal-shunt responses in 47 disjoint v661 cells from the same mouse. Every dictionary has one common broadcast and seven spatial columns. **C**, Paired ancestry advantages in residual energy capture after removing broadcast. Random routes sample anatomical sites; shuffling preserves column nonzero counts; surrogates preserve segment depths and parent out-degrees. **D**, Separate original, v661 and Pinky cohorts. Filled/open symbols indicate total/residual capture; Pinky's eight eligible cells contain only 9–13 excitatory-bearing sites each. **E**, Actual rank and nonzero density relative to dense eight-column wiring in v661. Equal column count does not ensure equal rank or wiring; ancestry and shuffling have identical nonzero counts. Randomized controls average 200 draws per cell. Symbols and bars show cell means and 95% bootstrap intervals; these describe within-cohort consistency. Capture uses squared excitatory-area-weighted field energy. The common-constrained SVD is an oracle capacity ceiling, and modeled fields are not observed teaching signals.

393 The column $G^{-1}\mathbf{e}_k$ gives the voltage response to unit current at k . On a passive tree with a somatic loss, the
 394 resulting adjoint change has an exact ancestry structure: all descendants of the shunt share the attenuation factor
 395 $[1 + \kappa_k(G^{-1})_{kk}]^{-1}$, while each sister block along the path to the soma shares its own factor (Fig. 8A; SI S4). These
 396 disjoint blocks form an ancestry partition. Weighting their indicators by the baseline soma-to-site transfer gives a
 397 dictionary on which the shunt acts through diagonal gains, providing a physical instance of a diagonal channel-gain
 398 matrix, denoted Γ_u . The diagonal form applies in this partition basis; it need not hold for an arbitrary selected set of
 399 overlapping routes.

400 In the small-perturbation fields used for Fig. 7, the driving-force term was substantial: its weighted squared
 401 energy ranged from 0.99 to 3.50 times that of the full field across the eight initial cells. It opposed the adjoint term
 402 (weighted correlations -0.979 to -0.913). Despite this cancellation, 97.1–99.9% of full-field energy remained
 403 within each focal ancestry partition. The small outside-partition residual therefore does not imply a negligible
 404 driving-force contribution (Supplementary Section S14).

405 We compared descendants $D(k)$ with off-route sites $C(k)$ at similar depth. A separate kernel criterion tests
 406 whether the adjoint change is larger below the shunt (Methods); within-cell, reversal-potential and contact-label
 407 controls are in Supplementary Fig. S21. The analyzed gradient was sensitivity to open excitatory conductance,
 408 $\gamma_i = q_i(E_E - V_i)$. The local driving force $E_E - V_i$ can change differently within each block, so the complete synaptic
 409 gradient need not share a single block gain. With primes denoting post-intervention values and ε_γ a numerical
 410 floor defined in Methods, we measured absolute log change, $m_i = |\log[(|\gamma'_i| + \varepsilon_\gamma)/(|\gamma_i| + \varepsilon_\gamma)]|$, and localization
 411 $\Lambda_k = \text{median}_{D(k)} m_i - \text{median}_{C(k)} m_i$. Positive localization means a larger gradient change below the shunt.

412 In the permissive normalized model, shunts changed descendant gradients more strongly than depth-matched
 413 off-route gradients, with increasing localization at higher dose (Fig. 8B; Supplementary Fig. S48). Current injection
 414 changed descendant gradients roughly threefold less. Its off-route changes were zero by construction under the
 415 soma-restoring control; this is not a general claim that current injection cannot change a gradient. Holding the
 416 post-shunt voltage but replacing its adjoint with the baseline adjoint removed most localization, supporting a transport
 417 contribution beyond the voltage change alone (Fig. 8C).

418 Physical calibration constrained the spatial selectivity of the gradient change. At axial resistivity $R_a = 150\Omega\text{cm}$
 419 and specific membrane resistance $R_m = 15,000\Omega\text{cm}^2$, mean shunt-minus-current localization was small and opposite-
 420 signed in the initial eight cells and a disjoint 45-cell cohort from the same mouse: -0.0019 and $+0.0024$, respectively.
 421 Signed changes showed attenuation of both descendant and off-route gradients at high resistance; lowering resistance
 422 primarily increased descendant attenuation (Fig. 8D). This increased the localization contrast by more than an order
 423 of magnitude (Fig. 8E). Distributed background conductance produced the same qualitative recovery at fixed high
 424 membrane resistance (Supplementary Fig. S11), consistent with a dependence on electrotonic state [25, 43].

425 Adding weak voltage-dependent channel currents produced a dose response close to the passive response at
 426 $R_m = 1,000\Omega\text{cm}^2$ with one baseline-leak unit of distributed background conductance (SI S9). This checks the local
 427 linearization around solved steady states. The baseline Jacobian included gate derivatives but remained fixed across
 428 doses, so the check does not establish robustness to strongly active dendrites or channel kinetics.

429 Together, the exact partition identity and the electrical calibration connect the morphology-defined dictionary
 430 to conductance-dependent route gains [26]. Their spatial selectivity depends on cable state. These imposed, state-
 431 matched perturbations measure modeled transport, rather than endogenous error production, observed plasticity or a
 432 learning advantage.

433 Measured responses test ancestry alignment and expose a transfer-geometry limit

434 Measured responses ask whether functional relationships preferentially follow ancestry. Seven MICrONS targets
 435 had complete arbors and imaged partners: 69 partners in the selected-scan dataset and thirteen eligible scans in the
 436 scan-complete analysis. These are recordings during stimulus presentation, not learning.

437 The direct structure–function test asked whether partners with more shared dendritic ancestry had more similar
 438 condition-mean responses, after controlling Euclidean separation and soma-to-contact path-length differences. Mean
 439 partial rank correlations were -0.069 for selected scans and -0.048 for all eligible scans; both target-bootstrap
 440 intervals included zero (Fig. 9A). This empirical test provides no support for preferential ancestry alignment in the

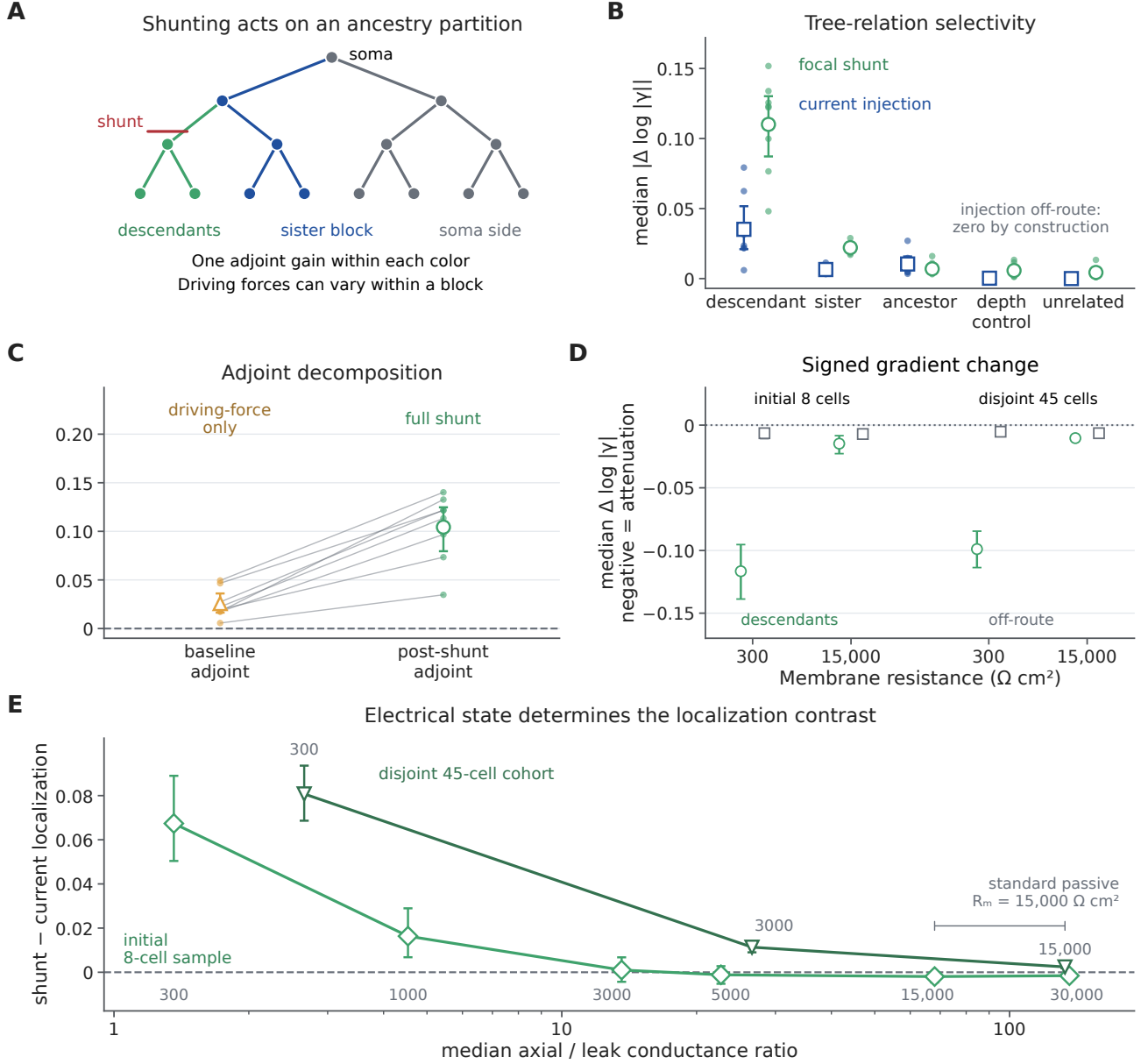


Figure 8: Focal shunts set ancestry-defined gains whose selectivity depends on electrical state. **A**, On a passive tree with a fixed somatic loss derivative, a shunt gives the same adjoint attenuation factor to every site in each colored ancestry block. The resulting gain is diagonal in the baseline-weighted partition dictionary; local synaptic driving forces can still vary within a block. **B**, Absolute log change in excitatory-conductance gradient magnitude by tree relation at unit dose, in eight reconstructed cells under permissive normalized passive parameters. A current injection matches baseline focal current, and a separately solved somatic current restores baseline somatic voltage after either intervention. Off-route injection zeros arise from this construction. **C**, Keeping the post-shunt voltage while restoring the baseline adjoint removes most localization. Lines pair cells. **D**, Signed log change in gradient magnitude at the two physical-calibration endpoints, with $R_a = 150 \Omega \text{ cm}$ and $R_m = 300$ or $15,000 \Omega \text{ cm}^2$. Circles and squares show descendants and depth-matched off-route sites; negative values denote attenuation. Site medians are averaged within cells; symbols and whiskers are cohort means and 95% whole-cell bootstrap intervals (20,000 draws). **E**, Electrical calibration in the initial eight cells and a disjoint 45-cell cohort from the same mouse. The horizontal axis is the cohort-median axial/leak ratio at fixed axial resistivity $R_a = 150 \Omega \text{ cm}$; numerical labels give membrane resistance R_m in $\Omega \text{ cm}^2$. Shunt-minus-current contrasts are small and opposite-signed at $R_m = 15,000$, and increase at lower resistance. Symbols and whiskers are cohort means and 95% cell-bootstrap intervals after averaging sites within cell. These state-matched perturbations measure modeled transport, not plasticity or learning benefit.

441 sampled partners. Its scale differs from population-level like-to-like wiring [44].

442 A response-level sensitivity simulation preserved observed partners, shared inputs, repeated stimuli and seven-
 443 target inference while calibrating noise to measured repeat reliabilities. Under its specified Gaussian ancestry-signal
 444 model, 80% detection corresponded to a mean simulated target-level partial rank correlation of approximately 0.25
 445 (Supplementary Fig. S56). This conditional estimate does not exclude weaker effects, other signal geometries or
 446 alignment involving unobserved partners.

447 A separate offline learning comparison used presynaptic responses to predict each target’s response through
 448 its complete passive tree. Trainable nonnegative excitatory conductances received input proxies at the partners’
 449 dominant mapped contact segments; a linear somatic readout predicted the target. Stimulus identities were held
 450 out, and transformations and model choices used training data only. Exact learning supplied trial-specific spatial
 451 sensitivities. Each restricted rule instead projected the baseline soma-to-site transfer profile $\mathbf{v}_0$ into a four-route
 452 dictionary D , giving $\mathbf{v}_D = P_D \mathbf{v}_0$, and delivered

$$\hat{\delta}_{i,D}^V = \delta_i^{\text{out}} \mathbf{v}_D. \quad (9)$$

453 Here δ_i^{out} is the scalar output error on trial i . The readout and bias always received exact gradients; only input-
 454 conductance credit differed.

455 The realized dictionaries explain what this comparison can test. Selected subtree routes were predominantly
 456 distal supports: across scans, they reached 48.0% of mapped partner inputs and contained an average of 1.31
 457 inputs per route. Several inputs can share one physical segment. Six of thirteen scans had only single-input routes
 458 (Fig. 9B,C). Moreover, one unrestricted fixed baseline-transfer profile already reconstructed 98.4% of the exact
 459 field and 97.6% of exact updates. The model therefore offered little trial-dependent spatial variation for a richer
 460 encoder to exploit. Its learning comparison chiefly diagnoses loss of a nearly fixed transfer profile when that profile
 461 is restricted to incompletely covering supports.

462 Consistent with that geometry, the subtree rule had no reliable held-out advantage over random anatomical
 463 routes or site shuffles. Mean NMSE was 0.788 for exact learning, 0.803 for ridge and 0.832 for restricted ancestry
 464 (Fig. 9D). Exact learning improved over ridge by 0.015 (95% target-bootstrap interval, 0.004–0.027), whereas the
 465 ancestry-minus-ridge difference was uncertain (0.029, -0.013 to 0.074). Here NMSE is relative to predicting the
 466 training mean, unlike population-variance normalization in the artificial tasks. Ten split/initialization replicates were
 467 averaged within scans and then within targets; the seven targets were the inferential units.

468 Allowing optimal trialwise coefficients within the same dictionaries barely improved ancestry update reconstruc-
 469 tion (Fig. 9E). For trial-by-parameter update matrices U , we measured

$$\mathcal{M}_\Delta(D) = 1 - \frac{\|U_{\text{exact}} - U_D\|_F^2}{\|U_{\text{exact}}\|_F^2}, \quad (10)$$

470 where the squared Frobenius norm sums squared entries. Ancestry rose from 0.232 for its fixed profile to 0.244 with
 471 oracle coefficients, compared with 0.385 for random routes and 0.468 for shuffles. Thus the primary limitation was
 472 the selected span’s geometry, not coefficient freezing. The re-executed checkpoint discrepancies and exact oracle
 473 definitions remain in Methods and Supplementary Table S9.

474 Input-removal and noise controls test dependence on recorded partners (Supplementary Fig. S34); imposed-
 475 alignment fields establish learning capacity when targets lie in the anatomical span (Supplementary Fig. S5). Neither
 476 tests endogenous teaching. The empirical ancestry-similarity test and modeled transfer diagnostic therefore bound
 477 distinct biological claims.

478 Discussion

479 Dendritic trees provide spatial addresses through which learning can distinguish conflicting synaptic demands. Which
 480 distinctions are useful depends on the task and the local sensitivity already available to the learner. Morphology
 481 supplies delivery patterns, teaching mechanisms supply their coefficients, and conductance regulates their gains.
 482 Image learning benefits mainly from preserving neuronal identity, including with random between-neuron feedback.

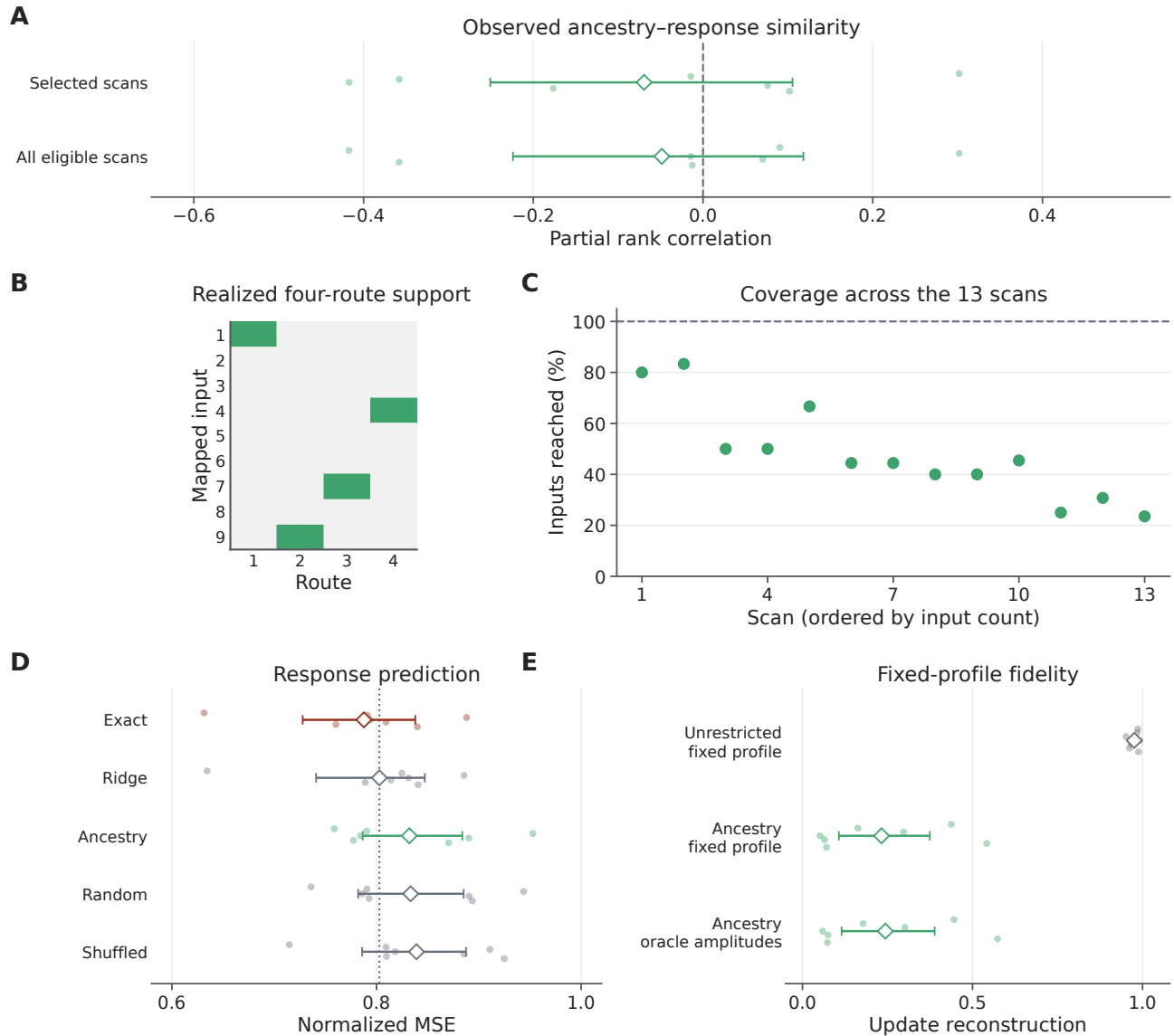


Figure 9: Observed ancestry alignment and a separate transfer-geometry diagnostic. **A**, Target-level partial rank correlations between shared ancestry and response similarity after controlling Euclidean separation and contact-depth differences. Selected scans contain 69 partners; all eligible scans comprise thirteen scans of the same seven targets. **B**, Actual input-by-route support in the scan with median mapped-input count, chosen without response outcomes; green entries indicate nonzero routes. Input-segment and route identifiers are retained in Source Data; several inputs can share a segment. **C**, Fraction of mapped inputs reached by selected ancestry routes in each scan, ordered by input count; scans are descriptive repeated observations within targets. **D**, Held-out normalized MSE for exact and restricted nonlinear learning and training-tuned ridge regression. NMSE uses the training-mean predictor as baseline. **E**, Common-checkpoint update reconstruction by the unrestricted fixed transfer profile, its restriction to ancestry routes, and optimal trialwise amplitudes within that ancestry span. A value of one is exact reconstruction. The last two differ in coefficient access, not in spatial support. In **A,D,E**, dots are seven target values; diamonds and whiskers show means and 95% target-bootstrap intervals. Panels **B–E** describe an offline model comparison and do not measure biological learning signals.

Contextual conflict favors branch selection, supplied through feedback or local eligibility gating. Hierarchical distractors favor ancestry at a bandwidth predicted by the generator's coefficients. These results distinguish useful information from its physical carrier.

The interaction comparison shows why input rank alone is insufficient. Pairwise and quartic targets share input-sensitivity spectra and a compatible tree but develop different credit spectra and tolerance of fixed profiles, despite initial calibration and common-rate controls. Spatial direction matters alongside dimension: even a nearly one-dimensional field can be poorly captured by uniform broadcast. These are learned-state properties, not retrospective support for the failed initialization selector. Intermediate dictionaries, adaptive profiles and alternative dynamics remain plausible solutions.

Conductance trees connect this distinction to a dendritic computation. Simple credit learns some compatible teachers accurately, whereas opposing branch tuning under contextual inhibition benefits from branch selection. A local inhibitory gate supplies that selection without oracle path coefficients, provided proximal credit remains ungated. The task supplies the inhibitory context and somatic error; discovering these signals remains a separate problem. Forward compatibility remains essential: the scalar multi-affine cut criterion connects input grouping to hierarchical tensor descriptions and supports structure estimation from noisy examples [45–47]. Serial composition likewise helps remove distributed nuisance factors, but the LocalCA accuracy advantage of exact credit changes sign between the 180- and 600-epoch budgets while exact credit retains lower cross-entropy. The gap is still moving at the cap. Task rank, physical depth and credit resolution therefore describe different constraints.

Obtaining coefficients remains a limitation beyond this explicit context gate. In the separate soft-encoder experiment, the prespecified noisy-cue condition remained 54.47 percentage points below oracle routing (Supplementary Fig. S32). Even with sixteen noise-free calibration trials, the encoder identified every context yet achieved only 19.4% task accuracy; exploratory maximum-probability readout reached the 80.3% oracle accuracy. This isolates leakage into incompatible routes as a failure mechanism. The exploratory rescue qualifies the primary deficit without replacing it. Exact paths and projected coefficients elsewhere remain information-access controls, and successful routing requires both an adequate span and coefficients that preserve its useful distinctions.

Reconstructed arbors compress their modeled passive response fields beyond a common broadcast. Degree–depth surrogates explain much of this capacity, and the target fields themselves favor ancestry through passive transport under a somatic loss. Testing alignment beyond that construction requires measured fields or a model with different error sources or active dynamics; changing the passive perturbation family alone would not suffice. Focal shunting supplies a biophysical link by reweighting the baseline-weighted ancestry partition. Driving-force changes partly cancel adjoint changes, while cable state sets the spatial selectivity of the resulting gradient. This concerns sensitivity transport, distinct from inhibition that computes errors or regulates forward gain and load [48–51].

Measured ancestry–response similarity provided no evidence of preferential alignment in the small recorded cohort. The offline learning comparison chiefly tests preservation of an almost fixed transfer profile and offers limited evidence about endogenous route use. A direct experiment would compare plasticity below a focal inhibitory site with electrotonically matched off-route sites while matching somatic voltage, first-order current and outcome signal. Changing conductance state would test the predicted separation between shared and ancestry-selective effects [22, 52, 53].

The framework complements somatic prediction, segregated teaching, interneuron-mediated errors and burst plasticity [32, 34–36], alongside cable-learning and engineered gating models [39, 54–57]. Spikes, temporal eligibility, recurrence and channel kinetics may change the available profiles and coefficient estimation [58, 59]. The organizing question remains which task-relevant distinctions the arbor and learning mechanism can jointly preserve.

Methods

The Methods define the models, task manipulations and inferential units behind the main comparisons. The Supplementary Information's contents and cohort index (Table S36) locate derivations, full protocols and historical cohorts. Sections S1–S4 cover theory and interactions, S5 artificial-network protocols, and S6–S12 anatomy, perturbations and measured responses. S14 details local-gate generation, budget sensitivity and measurement power.

Resolved configurations, seeds, checkpoints, exclusions and source hashes accompany the numerical tables. Reusing a cohort for a diagnostic does not add independent observations.

Conductance models and delivery coordinates

In a directed tree, $[b_1, \dots, b_D]$ lists branching factors from soma to tips, giving $\sum_{d=1}^D \prod_{r=1}^d b_r$ nonsomatic compartments. A child c supplies output $a_c = f_c(V_c)$. With leak g_n^L , aggregate open excitatory/inhibitory conductances $G_n^X = \sum_{i \in \mathcal{J}_n^X} g_i^X x_i^X$ for contact class X , and child couplings $g_{c \rightarrow n}^{\text{den}}$, the steady voltage is

$$V_n = R_n^{\text{tot}} \left(g_n^L E_L + G_n^E E_E + G_n^I E_I + \sum_c g_{c \rightarrow n}^{\text{den}} a_c \right), \quad (R_n^{\text{tot}})^{-1} = g_n^L + G_n^E + G_n^I + \sum_c g_{c \rightarrow n}^{\text{den}}. \quad (11)$$

Here $x_i \geq 0$ is contact activity, $g_i \geq 0$ its conductance, and E_X its reversal potential. A terminal-to-soma pass evaluates the directed steady state. Artificial conductance trees use dimensionless leak one and reversals $E_L = E_I = 0$, $E_E = 1$. Monotone parameter transforms enforce nonnegative conductances. Branch-output nonlinearities, termed reactivations in the implementation, are identity functions when disabled. Both excitatory and inhibitory contacts learn in the regular-tree experiments.

For a reciprocal cable residual $\mathbf{F}(\mathbf{V}, \mathbf{g}, \mathbf{x}) = 0$, the nonsingular voltage Jacobian J_V gives the adjoint

$$J_V^T \mathbf{q} = \nabla_{\mathbf{V}} \mathcal{L}, \quad \partial \mathcal{L} / \partial g_i = -\mathbf{q}^T \partial \mathbf{F} / \partial g_i. \quad (12)$$

With passive current balance, the synaptic gradient is $x_i(E_i^{\text{rev}} - V_n)q_n$. This established sensitivity calculation supplies the reference for restricted credit [9, 10, 33, 39]. Directed LocalCA combines local eligibility with delivered credit. For output $a_n = f_n(V_n)$, activation error $\delta_n^a = \partial \mathcal{L} / \partial a_n$ gives voltage error $\delta_n^V = f'_n(V_n) \delta_n^a$. The image implementation retains f'_n in local eligibility and replaces δ_n^a ; its gate updates also retain their exact local derivatives. Thus a fixed activation-space profile can correspond to a state-dependent voltage-space profile. Supplying the exact field recovers the exact selected gradients.

The raw-additive control retains the graph but sums signed inputs without the conductance denominator or reversal-potential driving force. Historical normalized-additive controls additionally standardize voltage across stage coordinates within each example; Supplementary Figs. S1–S4 identify those variants. The optional four- and five-factor rules add stage-level preconditioners and are described in Supplementary Section S2.

Writing a delivery dictionary as $A\Gamma$ separates spatial profiles in A from diagonal channel gains in Γ . Nonzero gains preserve the column span and oracle capture, while changing delivered coefficients and conditioning. A passive focal shunt admits this representation in its baseline-weighted ancestry partition; arbitrary overlapping dictionaries need not. Physical stage count, feedback rank, external error dimension and state-dependent path resolution are distinct resources (Supplementary Tables S13–S14).

Image feedback ladders and checkpoint diagnostics

For example i , the trained linear decoder supplies activation errors $\delta_{i,u} = \partial \mathcal{L} / \partial a_{i,0,u}$ at the somatic outputs of neurons u . The strict scalar is their mean unless its magnitude is at most 10^{-6} , in which case it is the largest-magnitude coordinate, preserving that coordinate’s sign. The same scalar reaches every core compartment, including the somatic gate. Neuron-specific feedback repeats each neuron’s own coordinate throughout its arbor; exact transport multiplies it by the local path derivatives. The historical matched-width fallback retained neuronal identity at the 128-wide somatic stage, affecting 640 of 2,413,312 core parameters. A separately trained strict-scalar control tests that exception (Supplementary Table S16).

The new MNIST ladder uses 128-soma [3, 3] shunting and raw-additive networks, with 40 excitatory and 20 inhibitory contacts per nonsomatic compartment. Within each architecture, rules share initial contact construction, initialization policy, decoder and data streams. Pixels are divided by 255 and flattened without further normalization or augmentation. Each seed divides the 60,000 training images into 48,000 training and 12,000 validation examples; all 10,000 official test images remain held out. Training uses 180 epochs, batches of 256 and the minimum-validation-cross-entropy checkpoint, without early stopping. Adam uses $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\varepsilon = 10^{-8}$ and no weight decay.

Original rates are 0.0015 for input synapses and decoder, 0.0007 for branch couplings and 0.0005 for reactivation gates.

Projected $K = 1$ averages all twelve nonsomatic activation errors. The $K = 3$ dictionary instead averages each proximal compartment with its three distal children. Both use the exact field and preserve its mean at identical weights; their norms can differ. Both projected rules retain exact somatic errors and local activation derivatives. A decoder-only reference trains the readout on the initialized core, whose unchanged hash is verified. Three development seeds compare rate multipliers 0.3, 1 and 3 for every rule and architecture, giving 108 fits. Validation alone selects rates. Ten fresh paired seeds per architecture and rule give 120 selected-rate and 120 original-common-rate outcomes from 190 unique fits; 50 fits serve both policies. The projected- $K = 1$ addendum was frozen before fresh outcomes. Complete protocols, clipping and excluded implementation checks are in Supplementary Section S5.

The original fifteen-seed MNIST ladder remains in Supplementary Fig. S45. Fashion-MNIST uses ten paired seeds; the raw-additive CIFAR-10 ladder uses twenty and a 20-soma $[3, 3, 3, 3]$ tree. Their distinct protocols and initialization are retained in SI. The separate fifteen-seed direct-feedback-alignment factorial replaces the decoder derivative by a fixed Gaussian output-to-neuron matrix, scaled by inverse square root of output dimension. It pairs random-feedback fits with the original exact-readout reference. Ownership controls permute neuronal recipients without changing coordinate values or bandwidth.

Common-checkpoint diagnostics compare rules at identical weights and states. Exact path transport is checked against automatic differentiation. New capture measurements use 2,048 held-out images at initial and exact-trained states; activation- and voltage-space metrics are kept separate. Gradient cosine, error magnitude by depth and mean-subtracted path energy are distinct summaries; none alone certifies learning performance. Historical depth and noise cohorts, including their task-identity and conductance-validity records, are specified in Supplementary Section S5.

Contextual conflict and ancestry routing

In the branch-conflict task, one context-selected branch determines a logistic output while all B branches form ungated local eligibilities. Nonselected Fashion-MNIST views are replaced independently by opposite-class images with probability χ . Exact feedback is $\delta_i \mathbf{1}[b = b_i^*]$; shared feedback is δ_i/B , where δ_i is output error and b_i^* the selected branch. Twenty paired seeds cross $B = 2, 4, 8$ with eight conflict doses. The algebraically equivalent gated-point calculation and BP receive the same updates; cyclic derangement preserves rank and support while changing the recipient.

Under centered, symmetric class evidence, shared-mode compatibility is $s(\chi) = 1 - 2\chi(B - 1)/B$, with zero at $\chi_c = B/[2(B - 1)]$. This mean-field prediction is separate from a trained accuracy crossing. The latter is reported both by interpolation of the mean curve and by each seed’s first sampled at-or-below-chance dose, retaining non-crossings. Checkpoint re-execution reproduces endpoints and compares exact/shared fields at both common and own trained states (Supplementary Fig. S29).

The ancestry factorial contains eight eight-dimensional blocks. Context c_i and binary label sign s_i are balanced, and inputs are $\mathbf{x}_{i,b} = a(b, c_i)s_i \mathbf{t}_b + \boldsymbol{\epsilon}_{i,b}$, with seed-specific unit teachers $\mathbf{t}_b$ and Gaussian noise $\boldsymbol{\epsilon}_{i,b} \sim \mathcal{N}(0, 1.1^2 I_8)$. The selected coefficient is one; sibling, same-half nonsibling and opposite-half coefficients are $-0.15, -0.45, -0.75$. A leaf permutation changes task-to-route alignment without changing the streams. For budget K , ancestry feedback has context-by-block entries $\Phi_{cb}^{(K)} = \sqrt{K/8} \mathbf{1}[g_K(\pi(c)) = g_K(b)]$, where g_K indexes contiguous groups and π maps streams to leaves. Thus every row has unit Euclidean norm.

The update is $\Delta \mathbf{w}_b = -(8\eta/N) \sum_i \delta_i \Phi_{c,b}^{(K)} \mathbf{x}_{i,\pi^{-1}(b)}$, with N training examples and rate η . Each seed supplies 2,048 training, 2,048 test and 1,024 switch examples. Full-batch descent uses rate 0.3, eighty training epochs and twenty switch epochs. Controls include deranged ancestry, depth-interleaved and random-sparse supports, dense rank- K fields and a training-informed PCA reference. The best matched control is the per-seed post-training maximum over the four non-PCA controls. Integer test counts preserve ties in the named Wilcoxon comparison; Holm correction covers four budgets. Alternative mean-sign-flip and historical BH analyses remain explicitly labeled in Supplementary Table S2. The separate noisy-context encoder and cue-delay experiments are fully specified in Supplementary Section S5.

619 Update utility and prospective selection

620 Let the exact gradient be $\boldsymbol{\mu}$ and the delivered update estimator have mean $\boldsymbol{m}$ and covariance Σ_{route} at the current
 621 weights. For an L -smooth loss and a nonnegative step η , expected loss after the update is bounded above by $\mathcal{L} -$
 622 $\eta \boldsymbol{\mu}^\top \boldsymbol{m} + (L\eta^2/2)(\|\boldsymbol{m}\|^2 + \text{tr}\Sigma_{\text{route}})$. Optimizing this expression gives utility $U = [\boldsymbol{\mu}^\top \boldsymbol{m}]_+^2 / [2L(\|\boldsymbol{m}\|^2 + \text{tr}\Sigma_{\text{route}})]$,
 623 with zero for a zero denominator. Utility certifies one-step progress under the stated moments; it is not a long-horizon
 624 ordering.

625 Synthetic quadratic screens isolate signal retention, noise rejection and fixed gains over sixteen Haar coordinates,
 626 without a forward neuron (Supplementary Fig. S46). Their matched signal/noise hierarchy is a designed crossover.
 627 The trained factorial’s retrospective utility uses actual full-batch updates and a global logistic-curvature bound;
 628 choosing maximum bandwidth matches its selection success. The separate prospective linear experiment retains
 629 all 12,800 fits, the initialization selector’s loss to fixed maximum-budget and rank-only baselines in every seed,
 630 and follow-ups forecasting evolving weights and noise. The 25,600-fit dynamics extension and observed-context-
 631 count control are reported in Supplementary Figs. S35 and S38–S39, with sealed choices, complete methods and
 632 computational costs.

633 Interaction compatibility, fixed profiles and learned spectra

634 For a subtree leaf set S , C_S arranges the target’s Fourier interaction coefficients by products inside and outside S ; C_S°
 635 removes the constant inside row. Scalar multi-affine trees represent the target exactly if and only if every centered
 636 cut has rank at most one. If $\mathcal{S}(T)$ contains all proper cuts of tree T and singular values are ordered from largest to
 637 smallest,

$$\text{MSE}(f, \hat{f}_T) \geq B(T; f) := \max_{S \in \mathcal{S}(T)} \sum_{j \geq 2} \sigma_j(C_S^\circ)^2. \quad (13)$$

638 Overlapping cut tails cannot be added as a lower bound. Supplementary Section S4 gives the constructive proof,
 639 minimum-depth algorithm and its relation to hierarchical tensor representations. The twelve-tree census, noisy-
 640 query estimator, end-to-end learning pipeline, conductance grouping controls and Boolean truth tables remain in
 641 Supplementary Figs. S36–S44. The theorem’s assumptions do not extend automatically to general conductance
 642 ancestors.

643 The matched credit experiment uses twenty fresh seed blocks after three development seeds select rates with
 644 equal tuning budgets. Matching and quartic conditions share tree, leaf assignment, initialization, inputs, additive
 645 noise and minibatch order. Exact, unit, initially calibrated and sign-only profiles use the same local eligibilities;
 646 calibration averages initial path derivatives over 256 unlabeled examples. Selected and common-rate outcomes are
 647 retained for SGD and Adam, giving 432 development and 1,200 fresh fits. Algebraic coefficients remain within
 648 $[-2, 2]$, and all bound contacts are recorded. The original fits stopped at 1,024 steps with seven saved checkpoints.
 649 A balanced extension retained all 720 unique algebraic trajectories at the selected/common-rate union, replayed
 650 every original checkpoint and continued to 16,384 steps. Its terminal and validation-selected results are descriptive
 651 analyses of the same seeds (Supplementary Section S14). Noisy held-out NMSE divides by clean target variance:
 652 one for matching and nested, one-half for quartic. All have noise variance 0.0225. The primary Adam contrast
 653 subtracts the pairwise calibrated-minus-exact gap from its quartic counterpart.

654 For evaluation-example-by-site path matrix Q , the best fixed profile is the leading right singular vector. Its capture
 655 is $\sigma_1(Q)^2 / \sum_j \sigma_j(Q)^2$, an oracle reconstruction with example-dependent amplitudes. Additional diagnostics evaluate
 656 uniform, calibrated and ancestry profiles, loss-residual weighting, eligibility-weighted updates and root-mean-square
 657 normalization of each site. Participation-ratio effective rank is $(\text{tr}C)^2 / \text{tr}(C^2)$ for uncentered $C = Q^\top Q / N$. At
 658 fixed weights and inputs, changing labels leaves the path field unchanged; eighty relabeling controls verify this.
 659 Independent checkpoint analysis and faithful replay of 600 earlier fits provide numerical and trajectory checks.
 660 Complete definitions, pairing audits, optimizer effects and the positive-conductance boundary are in Supplementary
 661 Section S4.

662 Conductance trees with context-dependent feature tuning

663 The seven-compartment model uses identity branch outputs in Eq. 11. Two independent $z_j \sim \mathcal{U}[-2, 2]$ generate
664 positive inputs $\exp(\pm z_j)$ for both subtrees. Every terminal has two excitatory and two inhibitory gains; binary
665 context inhibits one proximal compartment. All 24 conductances are positive and parameterized by logarithms.
666 Students start from terminal gains one, proximal inhibitory gains 15, daughter couplings 20 and 0.2 per subtree, and
667 soma couplings four. These proximal and coupling values match the teacher’s nominal scaffold, not its realized
668 weights. Independent Gaussian log perturbations have SD 0.4 in students and 0.15 in teachers. Full generators and
669 eligibility derivatives are given in Supplementary Section S4.

670 The local-gate follow-up used twenty new paired seeds and nine rules at Adam rates 0.01, 0.03 and 0.1, giving
671 1,080 trajectories. Each task supplied 2,048 training, 1,024 validation, 4,096 test and 512 unlabeled calibration
672 examples. The primary rate 0.03 was fixed from earlier work, not selected on these seeds. Minibatches contained 128
673 examples; logarithmic conductances stayed within $[-7, 7]$. Minimum validation NMSE selected states within fixed
674 4,096- and 16,384-update windows. The somatic error was $(V_s - y) / \text{Var}_{\text{train}}(y)$; NMSE uses clean target variance.
675 At a terminal with parent p , the hard gate multiplied this error by $\mathbf{1}[a_p = 0]$, where proximal inhibitory activity a_p
676 was zero or ten. Proximal and somatic multipliers remained one. The continuous gate used $1/(1 + g_p^I a_p)$. Both
677 combined these signals with the same exact local eligibilities and accessed no exact path derivatives.

678 Controls used exact credit, unit and initially calibrated broadcast, two weighted distal oracle profiles with unit
679 proximal credit, three oracle profiles including a proximal column, reversed distal gating, and gating at proximal
680 sites as well. Two primary paired sign-flip tests—opposed broadcast-minus-gate error and its opposed-minus-
681 aligned difference—were Holm-adjusted together. Other comparisons and 95% whole-seed bootstrap intervals were
682 descriptive. The protocol was committed after exposure to exploratory panel results but before execution on the
683 new seeds. Earlier development, wider-bound controls, gradient-cancellation analyses and all outcomes remain
684 in Supplementary Section S4 and Figs. S50–S53; the new rule definitions and validation are in Supplementary
685 Section S14.

686 Serial depth, sensor alignment and optimization

687 Physical-depth networks contain 64 somata and eight nonsomatic modules each. D1–D4 reallocate these modules
688 across branching vectors $[8]$, $[2, 3]$, $[2, 1, 2]$ and $[1, 1, 2, 2]$. Contacts and resources are matched; serial composition
689 changes. Grouped-point controls connect every module directly to the soma. Raw-additive controls retain the graph
690 without division, while flexible MLP controls are approximately parameter matched.

691 Excitatory blocks carry class evidence multiplied by lognormal nuisance factors. Inhibitory blocks sense those
692 factors with alignment α : zero removes trial-specific information while preserving the sensor distribution, and one
693 reports the factor. The generator floors noisy input rates before multiplying them by gains. Nested, flat and local-ratio
694 families distinguish distributed hierarchical support from generic factor cancellation and locally available ratios. Full
695 generators, floors, sensor-placement controls and contact inventories are defined in Supplementary Section S5.

696 Ten paired seeds support each original cohort. The initial maximum is 180 epochs with patience thirty; validation
697 selects model states. BP and LocalCA optimizer recipes differ in learning rates and parameter-group treatment, so
698 their endpoint gaps are reported separately. Exact-path LocalCA computes the exact selected gradients. An audit of
699 one hundred original trajectories verifies every configuration and final metric against its source: all fifty D3 fits across
700 the five relevant credit/optimizer arms reached the epoch cap with validation loss still falling. The original depth
701 contrasts are therefore fixed-budget outcomes. Sixty source-consistent restarts raise the maximum to 600 epochs,
702 retaining all fifty D3 fits and ten exact-BP D1 references. Within each restart, minimum validation loss selects states
703 in both budget windows. All fifty D3 fits and two D1 references reach the larger cap; eight D1 references stop early.
704 D3 validation loss continues to decrease, and its selected states are near the cap. Complete histories, accuracy and
705 loss contrasts, saved-state checks and source-concordance discrepancies are retained in Supplementary Section S5.

706 **Anatomical dictionaries and common-mode controls**

707 Observed MICrONS skeletons and mapped contacts define soma-rooted branch-to-branch segments. Initial eight
708 and disjoint 47-cell cohorts come from the same mouse; Pinky is a second mouse, with ten of twelve selected cells
709 meeting the inherited typed-contact criterion. Each subtree route contains the input-bearing descendants of its origin.
710 Contact labels are direct where available and otherwise proxy-based as specified in Supplementary Sections S6–S8.
711 Compression preserves path length but only approximates resistance; direct-label, mapping and geometry sensitivities
712 are retained.

713 The modeled passive perturbation field consists of reciprocal passive-cable gradient changes induced by small
714 focal shunts. Every corrected dictionary receives one constant column and $K - 1$ spatial columns. Ancestry selection
715 uses inhibitory fraction times excitatory-area-weighted descendant fraction without field outcomes. Depth bins
716 are reparameterized with the same constant while preserving their span; random routes, independent site shuffles
717 and degree–depth surrogates provide controls. The weighted common component is removed before computing
718 residual-energy capture. A common-constrained SVD oracle supplies the rank-budget ceiling. Numerical rank,
719 coverage and nonzero density are reported separately; equal column count alone does not match those resources.

720 The original eight cells are post hoc development for common-mode correction. The protocol was frozen before
721 evaluating the new independent-generator outcomes in v661 and Pinky. Two hundred randomized controls are
722 averaged within each cell. At eight columns, all 47 v661 cells and eight Pinky cells qualify; other eligible budgets are
723 retained without substituting for missing eight-column outcomes. Paired cell-bootstrap intervals and Holm-adjusted
724 comparisons describe within-cohort consistency, not animal-population replication. The older route-generated
725 positive controls use separate fields and endpoint definitions (Supplementary Section S7).

726 **Focal shunting and the electrotonic boundary**

727 Up to sixteen focal sites per cell span depth quartiles and require at least three excitatory descendants and three
728 comparison sites. Normalized E/I scales are 0.35 with reversals one and -0.2 . Shunt doses are 0.25, 0.5, one and
729 two times local membrane-plus-synaptic conductance. A matched injection supplies first-order current $\kappa_k(E_I - V_k)$
730 without changing the conductance matrix. A separately solved somatic current restores baseline voltage and is held
731 fixed during differentiation. The squared-loss target is one normalized unit below baseline; finite differences check
732 the adjoint gradients.

733 Localization subtracts depth-matched off-route from descendant medians of $|\log[(|\gamma'_i| + \epsilon)/(|\gamma_i| + \epsilon)]|$, with
734 $\epsilon = 10^{-15} \max(1, \max_i |\gamma_i|)$. Because this is unsigned, signed gradient changes are reported separately. Adjoint and
735 driving-force substitutions in $\gamma_i = q_i(E_E - V_i)$ distinguish transport from local voltage effects. Sites and relation-
736 reassignment templates are nested within cells. The passive ancestry-partition identity and its dictionary qualification
737 are derived in Supplementary Section S4.

738 Physical calibration uses cylindrical axial conductance $\pi r^2/(R_a \ell)$ and leak $2\pi r \ell/R_m$, expressed in nS, with
739 spherical soma area $4\pi r^2$. Here r and ℓ are segment radius and length, R_a axial resistivity and R_m specific membrane
740 resistance. Sweeps include $R_a = 100, 150, 250 \Omega \text{cm}$ and $R_m = 300\text{--}30,000 \Omega \text{cm}^2$, fixed and locally normalized
741 doses, and background conductance. The disjoint calibration cohort contains 235 eligible sites in 45 cells. Geometric
742 fallbacks and dose definitions are specified in Supplementary Section S9.

743 The weak-channel check uses $R_a = 150$, $R_m = 1,000$ in those units and background conductance equal to leak.
744 Median maximal Na, K, Ca and HCN conductances are 0.06, 0.10, 0.025 and 0.05 times leak; NMDA is 0.15 times
745 excitatory conductance. Sixty-four draws per cell passed the predefined steady-state checks. Responses use rank-one
746 updates of the fixed baseline Jacobian, including voltage-gate derivatives, followed by somatic state matching. They
747 are local linearizations rather than newly solved nonlinear equilibria at each dose (Supplementary Fig. S47).

748 **Measured-response prediction and geometric diagnostics**

749 The seven-target cohort includes thirteen eligible scans. Trial responses are the mean of the supplied ROI fluorescence
750 traces in `RoiResponseSeries` over each stimulus interval; our analysis did not deconvolve the traces or apply a
751 reliability correction. Repeated responses are averaged by stimulus condition for the ancestry-similarity analysis;

752 partial rank correlations control Euclidean separation and soma-to-contact path-length differences. Offline learning
753 holds out stimulus identities. Training-only percentile transforms produce nonnegative presynaptic proxies, and each
754 partner enters at its dominant mapped contact segment. A linear somatic readout predicts the standardized target
755 response. Only input-conductance credit differs across rules; readout and bias gradients remain exact.

756 Four-route dictionaries are selected without target responses and receive the projected baseline transfer profile, as
757 in Eq. 9. Realized supports are reconstructed directly from saved segment identifiers. The displayed matrix uses the
758 scan with median mapped-input count, with identifier-based tie breaking; all scan supports are retained. At common
759 exact-trained states, field and eligibility-weighted update reconstruction compare the fixed profile, per-trial field
760 projection and the best within-span update. These are oracle diagnostics rather than newly trained rules. The 130
761 exact re-executions retain their endpoint discrepancies.

762 Ridge regularization is selected within training data for every outer split, using the same measured predictors.
763 Observed-input removal, predictor noise and reliability masks are separate sensitivity analyses with no held-out
764 information used for selection. Ten split/initialization replicates are averaged within scans and then within seven
765 targets. Imposed-alignment quadratic fields and the six-animal external signed-credit reanalysis are separate datasets
766 and tests, fully described in Supplementary Sections S10–S12.

767 **Statistics and computational provenance**

768 Confirmatory choices precede outcome inspection but are not described as public preregistration. Pilots, post hoc
769 analyses and same-seed reruns are identified separately. Effect sizes accompany experiment-specific 95% intervals,
770 or adjusted 97.5% intervals where prespecified. Tests are two sided unless a directional protocol states otherwise.
771 Named multiplicity families and exact tie treatment are retained in Source Data; null results imply equivalence only
772 when a margin is tested.

773 For the sensitivity simulation, a positive-semidefinite shared-path covariance generated joint tuning across the
774 102 observed presynaptic roots. We mixed ancestry-correlated and independent reliable tuning, then added noise
775 calibrated to each recording’s split-half reliability and actual repeat counts. Four thousand paired datasets at each
776 of 21 signal fractions were analyzed with the original partial-rank statistic and an exact seven-target signed-rank
777 test. Detection required two-sided $P \leq 0.05$ and a positive mean effect; Wilson intervals describe Monte Carlo
778 uncertainty. The model and analysis were frozen before simulated outcomes. Supplementary Section S14 gives the
779 covariance, reliability calibration, dependence assumptions and exact test resolution; the released pair-level inputs
780 permit reproduction.

781 Independent units are paired training seeds, reconstructed cells, postsynaptic targets or external animals. Sites,
782 contacts, scans, splits, checkpoints and randomized draws remain nested. Seed counts reflect the experiment’s
783 purpose; no prospective parametric power calculation was performed. Source hashes certify retained inputs and
784 implementations, not unavailable historical runtime states. The matched credit and new conductance studies recorded
785 Python 3.10.13, NumPy 2.2.6 and pandas 2.3.3. The new MNIST controls used PyTorch 2.9.1/CUDA 12.8 and
786 torchvision 0.24.1 with NumPy 2.2.6; the depth extension used Python 3.10 and PyTorch 2.9.1. The isolated reviewer
787 environment was validated with NumPy 1.26.4, SciPy 1.15.3, PyTorch 2.9.1, torchvision 0.24.1 and Matplotlib
788 3.10.9. Cohort-specific runtime records distinguish executed experiments from this installation check. Supplementary
789 Section S13 lists all exclusions, statistical families, numerical checks and release records.

790 OpenAI Codex assisted with code, statistical and consistency audits, figure preparation, literature checks and
791 language editing. The authors remain responsible for design, interpretation, numerical accuracy, citations and the
792 final manuscript.

793 **Ethics and data reuse**

794 No new experiments involving animals or human participants were performed. All biological analyses reused public
795 MICrONS, DANDI, and published animal source data. The source studies describe the original data collection and
796 institutional approvals [22, 42, 60].

797 Data availability

798 MICrONS anatomy and connectivity are available through the public CAVE datastack minnie65_public. The
799 primary cohort used materialization 1822, skeleton-service reconstructions, and the synapses_pni_2 table. The
800 disjoint sensitivity cohort used the public v661 static release. Secondary target-structure labels used only in the
801 full-contact analysis came from the in-development synapse_target_predictions_ssa_v2 table. These labels
802 are distributed only as derived data and remain subject to the source-data access terms. All principal structural
803 findings were repeated without them.

804 The independent-animal replication used the MICrONS phase-1 layer-2/3 Pinky v185 fixed meshes, soma-
805 valence table, and dense synapse table archived at Zenodo DOI 10.5281/zenodo.3710459 [61]. File sizes, Zenodo
806 MD5 checksums, and local SHA-256 checksums are recorded in the supplied manifest. Volumetric imagery is
807 archived in BossDB under DOI 10.60533/BOSS-2021-T0SY, and co-registered functional data are available as
808 DANDI:000402. The supplied manifest records the exact asset and table identifiers, and the source publication
809 describes the full data release [42].

810 Public benchmark datasets are available from their original repositories. The inhibitory-census tables and SWCs
811 are available with the source record for Schneider-Mizell et al. [60]. The Francioni et al. workbook is available
812 through the Source Data link associated with the Nature article [22]; its checksum and the sheet names used here are
813 recorded in the manifest. Derived cell and exclusion manifests, numerical source data underlying every figure, and
814 non-restricted summaries are supplied with this manuscript. The measured-response release includes the analyzed
815 partner-pair tables, contact geometry, recording identifiers and repeated-stimulus response arrays needed for the
816 sensitivity analysis. Authentication tokens and externally hosted raw caches are not redistributed.

817 Code availability

818 The accompanying reviewer software archive contains the complete training implementation, exact-gradient and
819 perturbation analyses, versioned configurations, tests, figure scripts, and byte-identified snapshots of the critical
820 training components under the MIT License. A separate Source Data archive contains the numerical data underlying
821 every display item. These archives are available to editors and reviewers with the submission. The same version
822 will be released publicly under the MIT License in a versioned repository and permanent archival record upon
823 publication.

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Author contributions

H.S. developed the computational framework, performed the analyses, and wrote the initial manuscript. M.R. contributed to the theory of local credit assignment, its implementation, and its interpretation. B.L.S. supervised the work and contributed to the biological framing, study design, and manuscript. All authors reviewed and revised the manuscript.

Competing interests

The authors declare no competing interests.